

Chapter 19

Permission Control (PMS)

19.1 Overview

The permission control of ESP32-P4 consists of two parts: PMP (Physical Memory Protection) and APM (Access Permission Management).

PMP is located inside the HP CPUs and can control the HP CPUs' access to all address spaces. APM is an access permission management module located at the bus port, which can manage the permissions for the CPU (HP CPU0/1, LP CPU) to access the registers of all on-chip peripherals (see HP CPU PERI, HP PERI, and LP PERI in Table 19.1-1) and part of internal and external memory. The APM can also manage DMA masters like VDMA, USB OTG 2.0, to access parts of internal and external memory.

The APM module compares the configurable address range or the fixed address range, the access permissions of each address range, and the information carried on the bus like master ID, security mode, access address, access type, etc, to determine whether the access is allowed. With APM, users can precisely control the access of all masters to the memory and peripheral registers.

The distribution of management areas by PMP and APM is shown in Table 19.1-1. For HP CPU0/1, the control relationship between PMP and APM is shown in Figure 19.1-1.

Table 19.1-1. Management Areas by PMP and APM

Master Slave	HP CPU0	HP CPU1	LP CPU	DMA Masters ¹
HP ROM ²	PMP	PMP	HP APM	DMA APM
HP SPM	PMP	PMP	N/A	N/A
HP L2MEM ²	PMP	PMP	HP APM	DMA APM
EXT MEM ²	PMP	PMP	HP APM	DMA APM
Direct Access HP ROM ³	PMP + HP APM	PMP + HP APM	N/A	N/A
Direct Access HP L2MEM ³	PMP + HP APM	PMP + HP APM	N/A	N/A
Direct Access EXT MEM ³	PMP + HP APM	PMP + HP APM	N/A	N/A
HP CPU PERI ⁴	PMP + HP APM	PMP + HP APM	HP APM	N/A
HP PERI ⁵	PMP + HP APM	PMP + HP APM	HP APM	N/A
LP PERI ⁶	PMP + LP APM	PMP + LP APM	LP APM	N/A
LP ROM	N/A	N/A	N/A	N/A
LP SRAM	PMP + LP APM	PMP + LP APM	N/A	N/A

- ¹ DMA masters that support DMA transfers, such as VDMA, USB 2.0 OTG, MEM_MONITOR, etc.
- ²
- HP ROM (0x4FC0_0000 - 0x4FC1_FFFF)
 - HP L2MEM (0x4FF0_0000 - 0x4FFB_FFFF)
 - EXT MEM, including external flash (0x4000_0000 - 0x43FF_FFFF) and external RAM (0x4800_0000 - 0x4BFF_FFFF)
- ³ “Direct access” means HP CPU0/1 accesses internal and external memory mapped into the following address spaces without going through the cache:
- HP ROM (0x8FC0_0000 - 0x8FC1_FFFF)
 - HP L2MEM (0x8FF0_0000 - 0x8FFB_FFFF)
 - EXT MEM, including external flash (0x8000_0000 - 0x83FF_FFFF) and external RAM (0x8800_0000 - 0x8BFF_FFFF)
- ⁴ HP CPU peripheral registers (0x3FF0_0000 - 0x3FF1_FFFF)
- ⁵ HP system peripheral registers (0x5000_0000 - 0x500F_FFFF)
- ⁶ LP system peripheral registers (0x5011_0000 - 0x5012_FFFF)
- ⁷ As can be seen from the table, APM includes HP APM, LP APM, and DMA APM, which will be described in the following sections.

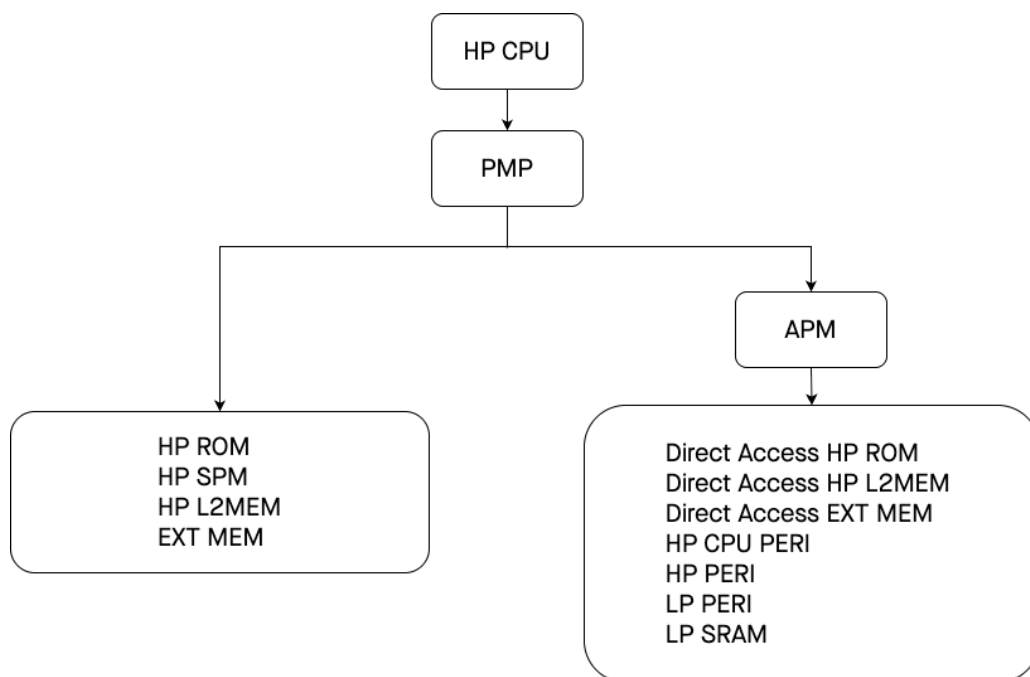


Figure 19.1-1. PMP-APM Management Relation

The diagram illustrates that when the HP CPUs access HP ROM, HP SPM, HP L2MEM, and EXT MEM, the access paths are solely managed by PMP. On the other hand, when the HP CPUs access the peripheral registers, HP ROM, HP L2MEM, and EXT MEM without going through the cache, the access paths are controlled by both PMP and APM. If the PMP check fails, the APM permission control will not be triggered.

PMP-related registers are located inside the HP CPUs and can be read or configured with special instructions. For how to configure PMP, please refer to Chapter 1 [High-Performance CPU > 1.10.1 Standard Physical Memory Protection](#).

The following sections will provide a detailed introduction to the features, functions, and configurations of the APM module. When referring to HP CPU0/1, accessing HP ROM/HP L2MEM/EXT MEM always means direct access to HP ROM/HP L2MEM/EXT MEM.

19.2 Features

The APM module has the following features:

- Up to 32 configurable address ranges for each DMA master
- Access permission management for each CPU core to access internal memory, external memory, and peripheral registers
- Support for interrupts
- Support for exception information record

19.3 Functional Description

19.3.1 Architecture

The APM module contains three parts and five register modules as follows:

- **DMA APM**, containing register group [HP_DMA_PMS_REG](#), responsible for:
 - configuring up to 32 address regions for DMA masters
 - managing the access permission to each address region for each DMA master
- **HP APM**, containing register groups [HP_PERI_PMS_REG](#) and [LP2HP_PERI_PMS_REG](#), responsible for managing the access permissions to peripheral registers (HP CPU PERI, HP PERI), internal memory (HP ROM, HP L2MEM), and external memory (EXT MEM) in the HP system, specifically,
 - managing the access permissions for HP CPU0/1 in user mode to access all the above-mentioned slaves
 - managing the access permissions for HP CPU0/1 in machine mode to access all the above-mentioned slaves
 - managing the access permissions for LP CPU in machine mode to access all the above-mentioned slaves
- **LP APM**, containing register groups [LP_PERI_PMS_REG](#) and [HP2LP_PERI_PMS_REG](#), responsible for managing the access permissions to peripheral registers (LP PERI) and internal memory (LP SRAM) in the LP system, specifically,
 - managing the access permissions for HP CPU0/1 in user mode to access LP PERI and LP SRAM
 - managing the access permissions for HP CPU0/1 in machine mode to access LP PERI and LP SRAM
 - managing the access permissions for LP CPU in machine mode to access LP PERI

Figure [19.3-1](#) shows the access paths managed by APM and the corresponding register groups.

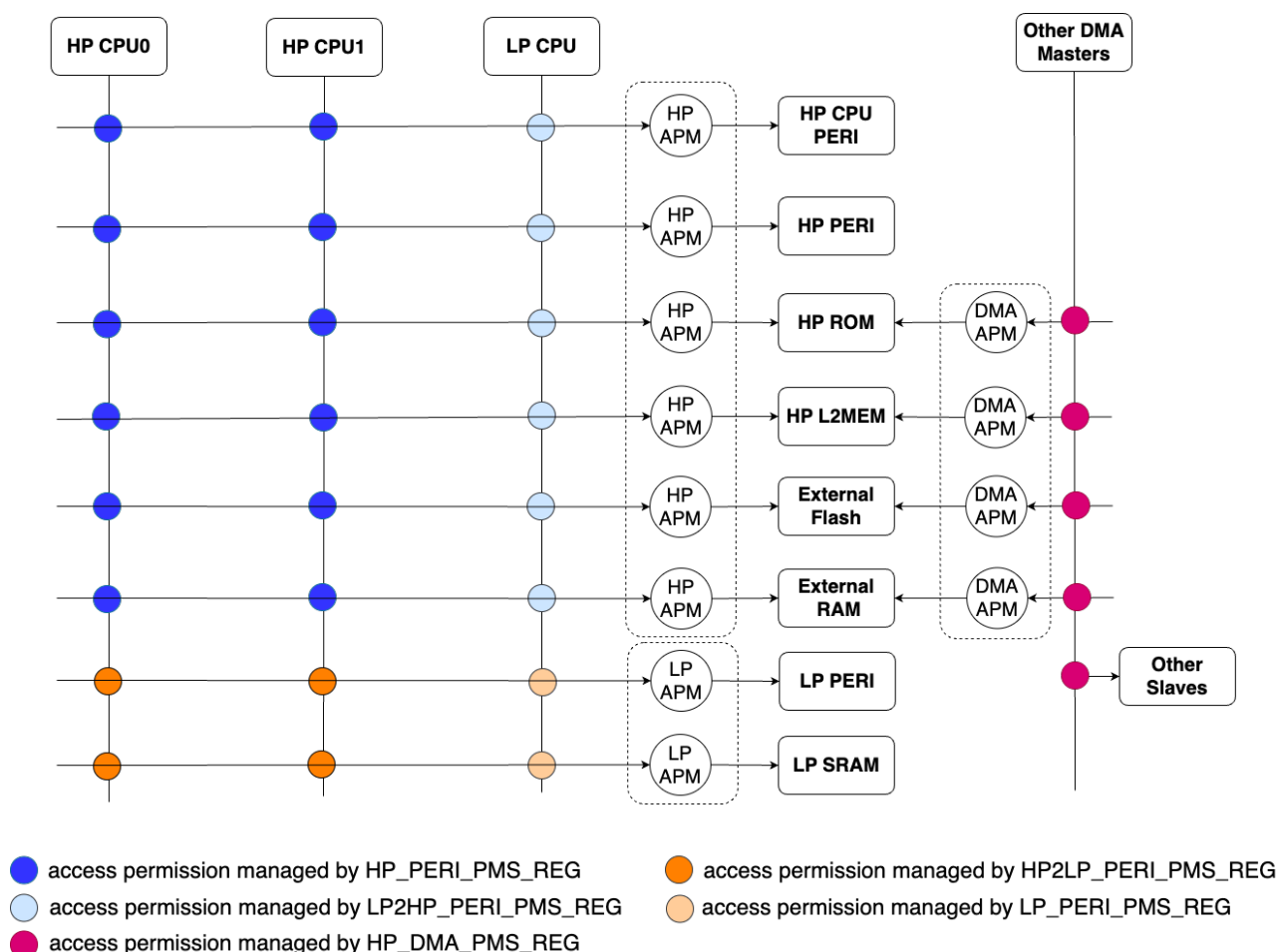


Figure 19.3-1. APM Architecture

Note:

- For register group names “HP2LP” and “LP2HP”, the “HP” refers to HP CPU0/1, and “LP” refers to LP CPU.
- From Table 19.1-1, it can be seen that HP CPU and LP CPU access different address spaces of HP ROM, HP L2MEM, external flash, and external RAM. For HP CPU0/1, HP APM manages their direct access to internal and external memory without going through the cache.
- [LP_PERI_PMS_REG](#) includes eight additional configurable address ranges for HP APM and LP APM to manage the access permissions to peripheral registers, including HP CPU PERI, HP PERI, and LP PERI.

19.3.2 Address Ranges and Permissions Management

APM manages address ranges, which can be fixed or configurable. DMA APM can manage up to 32 configurable address ranges. In addition to the fixed address ranges, HP APM and LP APM can also manage in total eight configurable address ranges.

19.3.2.1 Address Ranges Managed by HP APM and LP APM

In the access paths controlled by HP APM and LP APM, the internal memory and external memory use fixed address ranges for access control, while peripheral registers not only support fixed address ranges permission control but also support eight configurable address ranges.

- Fixed address ranges
 - The fixed address ranges are the valid address spaces of peripheral registers, internal memory, and external memory. See the table notes under Table 19.1-1 for the address ranges of internal and external memory. Refer to Chapter 7 *System and Memory* > Table 7.3-2 for the address ranges of each peripheral.
 - To manage access to the fixed address ranges, please refer to Figure 19.3-1 for the access paths and their corresponding register groups.
- Configurable Address Range
 - APM can configure eight address ranges with `PMS_PERI_REGION $n/m$ _LOW_REG` and `PMS_PERI_REGION $n/m$ _HIGH_REG` ($n/m = 0 \sim 7$) in `LP_PERI_PMS_REG`. These configurable address ranges can be used by HP CPUs and LP CPU to access all on-chip peripheral registers (HP CPU PERI, HP PERI, LP PERI).
 - The access permissions are managed by `PMS_PERI_REGION_PMS_REG` in `LP_PERI_PMS_REG`.

When managing access permissions to peripheral registers, the permission management for the configurable address range should precede that of the fixed address range. Specifically:

- If the access address is within the configurable address range, then APM will check if this access is permitted to access the configurable address range.
 - If permitted, APM will continue to check if the access is permitted to the fixed address range.
 - If not permitted, the access will be denied and skip the permission check to the fixed address range.
- If the access address is outside the configurable address range, then it will skip the configurable address range check, but will continue to the fixed address range check.
- The management of configurable address ranges is a supplement to the management of fixed address ranges. For example, users can configure access permissions for a fixed address range, and deny access to two configurable ranges within the fixed range, resulting in a fixed address range containing two access-disabled address ranges.

19.3.2.2 Address Ranges Managed by DMA APM

The register group `HP_DMA_PMS_REG` provides 32 sets of registers (`PMS_DMA_REGION $n$ _LOW_REG` ~ `PMS_DMA_REGION $n$ _HIGH_REG`, $n = 0 \sim 31$) to configure address ranges for DMA masters to access. Note that since the address must be aligned to 4K bytes, the lowest 12 bits are set to 0.

`HP_DMA_PMS_REG` provides separate read and write permissions for each DMA master to access each address range.

19.4 Programming Procedure

As HP CPUs can work in either user mode or machine mode, HP APM and LP APM grant specific access permissions for each work mode. Therefore, users need to configure the work mode for HP CPUs first in order for the HP CPUs to access internal memory, external memory, and peripheral registers.

Since LP CPU only supports machine mode, HP APM and LP APM grant access permissions for the machine mode of LP CPU.

DMA masters can access slaves regardless of the CPUs' work mode. DMA APM provides separate read and write permissions for each DMA master.

19.4.1 Configuring Access Permissions for HP CPU0/1

The configuration process for HP CPU0/1 accessing internal memory, external memory, or peripheral registers is as follows:

1. Configure user mode or machine mode for HP CPU0/1. For how to configure the work mode, please refer to RISC-V Instruction Set Manual, Volume II: Privileged Architecture, Version 1.10.
2. If accessing peripheral registers, eight additional address ranges can be configured, please refer to Section 19.3.2 for details.
3. Configure access permissions:
 - Configure the access permissions to internal memory (HP L2MEM and HP ROM), external memory (External RAM and flash), and peripheral registers (HP PERI and HP CPU PERI) in the HP system through `HP_PERI_PMS_REG`:
 - Configure the permission of HP CPU n accessing the `MODULE_NAME` module in `MODE_NAME` mode using `PMS_CORE $n$ _MODE_NAME_MODULE_NAME_ALLOW`, for example `PMS_CORE $n$ _MM_PSRAM_ALLOW`.
 - Configure the access permissions to LP SRAM and LP PERI through `HP2LP_PERI_PMS_REG`:
 - Configure the permission of HP CPU n accessing the `MODULE_NAME` module in `MODE_NAME` mode using `PMS_HP_CORE $n$ _MODE_NAME_MODULE_NAME_ALLOW`, for example `PMS_HP_CORE $n$ _MM_LP_SYSREG_ALLOW`.

19.4.2 Configuring Access Permissions for LP CPU

LP CPU only supports machine mode. The configuration process for LP CPU accessing internal memory, external memory, or peripheral registers is as follows:

1. If accessing peripheral registers, eight additional address ranges can be configured, please refer to Section 19.3.2 for details.
2. Configure access permissions:
 - Configure the access permissions to internal memory (HP L2MEM and HP ROM), external memory (External RAM and flash), and peripheral registers (HP PERI and HP CPU PERI) in the HP system through `LP2HP_PERI_PMS_REG`:
 - Configure the permission of LP CPU accessing the `MODULE_NAME` module in machine mode using `PMS_LP_MM_MODULE_NAME_ALLOW`, for example `PMS_LP_MM_PSRAM_ALLOW`.
 - Configure the access permissions to peripheral registers (LP PERI) in the LP system through `LP_PERI_PMS_REG`:
 - Configure the permission of LP CPU accessing the `MODULE_NAME` module in machine mode using `PMS_LP_MM_MODULE_NAME_ALLOW`, for example `PMS_LP_MM_LP_SYSREG_ALLOW`.

19.4.3 Configuring Access Permission for DMA Masters

The configuration process for DMA masters accessing internal memory, external memory, and other address spaces is as follows:

1. Configure the address ranges accessible to all DMA masters:
 - Configure the high 20 bits of the starting address of region *n* using `PMS_DMA_REGIONn_LOW_REG`.
 - Configure the high 20 bits of the ending address of region *n* using `PMS_DMA_REGIONn_HIGH_REG`.
2. Configure the read and write permissions for each DMA master (*DMA_MASTER*) to access each address range:
 - Configure the read permissions using `PMS_DMA_MASTER_PMS_R_REG`, such as `PMS_DMA_GMAC_PMS_R_REG`.
 - Configure the write permissions using `PMS_DMA_MASTER_PMS_W_REG`, such as `PMS_DMA_GMAC_PMS_W_REG`.

19.5 Illegal Access and Interrupts

When the APM module manages access paths, it parses the information carried on the bus (e.g., master ID, access direction (read or write), access address, and working mode) and then determines the access permission. When the APM module confirms that the access lacks permission, ESP32-P4 will consider it as an unauthorized access and will take the following actions:

- Reject the access request and provide an error response.
- HP APM will trigger `ICM_SYS_REG_INT` interrupt signal.
- DMA APM will trigger `ICM_SYS_REG_INT` interrupt signal.
- LP APM will trigger `LP_SYSREG_INT` interrupt signal.

These interrupt signals are generated by the internal interrupt sources of the APM module. Table 19.5-1 lists the internal interrupt sources, interrupt trigger conditions, and the generated interrupt signals.

Table 19.5-1. APM Internal Interrupt Sources

Internal Interrupt Source	Trigger Condition	Interrupt Signal
ICM_CPU_ADDRHOLE_INTR	Triggered when HP APM detects unauthorized access; or when DMA APM detects unauthorized access from high-speed USB 2.0 OTG, full-speed USB 2.0 OTG, EMAC, SDMMC, Trace0/1, GDMA-AHB, L2MEM Monitor, or SPM Monitor.	ICM_SYS_REG_INT
ICM_SYS_ADDRHOLE_INTR	Triggered when DMA APM detects unauthorized access from VDMA, GDMA-AXI, 2D-DMA, or H264 DMA.	ICM_SYS_REG_INT
LP_ADDRHOLE_INTR	Triggered when LP APM detects unauthorized access.	LP_SYSREG_INTR

APM automatically records relevant information about illegal accesses, including master ID, access direction (read or write), access address, illegal access flags, etc. All this information can be retrieved from the relevant registers described in Chapter 20 *System Registers (SYSREG)*.

19.6 Register Summary

19.6.1 HP_DMA_PMS_REG

The addresses in this section are relative to the HP_DMA_PMS base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Version Control Registers			
PMS_DMA_DATE_REG	Version control register	0x0000	R/W
Clock Gating Registers			
PMS_DMA_CLK_EN_REG	Clock gating register	0x0004	R/W
Region Configuration Registers			
PMS_DMA_REGION0_LOW_REG	Region0 start address configuration register	0x0008	R/W
PMS_DMA_REGION0_HIGH_REG	Region0 end address configuration register	0x000C	R/W
PMS_DMA_REGION1_LOW_REG	Region1 start address configuration register	0x0010	R/W
PMS_DMA_REGION1_HIGH_REG	Region1 end address configuration register	0x0014	R/W
PMS_DMA_REGION2_LOW_REG	Region2 start address configuration register	0x0018	R/W
PMS_DMA_REGION2_HIGH_REG	Region2 end address configuration register	0x001C	R/W
PMS_DMA_REGION3_LOW_REG	Region3 start address configuration register	0x0020	R/W
PMS_DMA_REGION3_HIGH_REG	Region3 end address configuration register	0x0024	R/W
PMS_DMA_REGION4_LOW_REG	Region4 start address configuration register	0x0028	R/W
PMS_DMA_REGION4_HIGH_REG	Region4 end address configuration register	0x002C	R/W
PMS_DMA_REGION5_LOW_REG	Region5 start address configuration register	0x0030	R/W
PMS_DMA_REGION5_HIGH_REG	Region5 end address configuration register	0x0034	R/W
PMS_DMA_REGION6_LOW_REG	Region6 start address configuration register	0x0038	R/W
PMS_DMA_REGION6_HIGH_REG	Region6 end address configuration register	0x003C	R/W
PMS_DMA_REGION7_LOW_REG	Region7 start address configuration register	0x0040	R/W

Name	Description	Address	Access
PMS_DMA_REGION7_HIGH_REG	Region7 end address configuration register	0x0044	R/W
PMS_DMA_REGION8_LOW_REG	Region8 start address configuration register	0x0048	R/W
PMS_DMA_REGION8_HIGH_REG	Region8 end address configuration register	0x004C	R/W
PMS_DMA_REGION9_LOW_REG	Region9 start address configuration register	0x0050	R/W
PMS_DMA_REGION9_HIGH_REG	Region9 end address configuration register	0x0054	R/W
PMS_DMA_REGION10_LOW_REG	Region10 start address configuration register	0x0058	R/W
PMS_DMA_REGION10_HIGH_REG	Region10 end address configuration register	0x005C	R/W
PMS_DMA_REGION11_LOW_REG	Region11 start address configuration register	0x0060	R/W
PMS_DMA_REGION11_HIGH_REG	Region11 end address configuration register	0x0064	R/W
PMS_DMA_REGION12_LOW_REG	Region12 start address configuration register	0x0068	R/W
PMS_DMA_REGION12_HIGH_REG	Region12 end address configuration register	0x006C	R/W
PMS_DMA_REGION13_LOW_REG	Region13 start address configuration register	0x0070	R/W
PMS_DMA_REGION13_HIGH_REG	Region13 end address configuration register	0x0074	R/W
PMS_DMA_REGION14_LOW_REG	Region14 start address configuration register	0x0078	R/W
PMS_DMA_REGION14_HIGH_REG	Region14 end address configuration register	0x007C	R/W
PMS_DMA_REGION15_LOW_REG	Region15 start address configuration register	0x0080	R/W
PMS_DMA_REGION15_HIGH_REG	Region15 end address configuration register	0x0084	R/W
PMS_DMA_REGION16_LOW_REG	Region16 start address configuration register	0x0088	R/W
PMS_DMA_REGION16_HIGH_REG	Region16 end address configuration register	0x008C	R/W
PMS_DMA_REGION17_LOW_REG	Region17 start address configuration register	0x0090	R/W
PMS_DMA_REGION17_HIGH_REG	Region17 end address configuration register	0x0094	R/W

Name	Description	Address	Access
PMS_DMA_REGION18_LOW_REG	Region18 start address configuration register	0x0098	R/W
PMS_DMA_REGION18_HIGH_REG	Region18 end address configuration register	0x009C	R/W
PMS_DMA_REGION19_LOW_REG	Region19 start address configuration register	0x00A0	R/W
PMS_DMA_REGION19_HIGH_REG	Region19 end address configuration register	0x00A4	R/W
PMS_DMA_REGION20_LOW_REG	Region20 start address configuration register	0x00A8	R/W
PMS_DMA_REGION20_HIGH_REG	Region20 end address configuration register	0x00AC	R/W
PMS_DMA_REGION21_LOW_REG	Region21 start address configuration register	0x00B0	R/W
PMS_DMA_REGION21_HIGH_REG	Region21 end address configuration register	0x00B4	R/W
PMS_DMA_REGION22_LOW_REG	Region22 start address configuration register	0x00B8	R/W
PMS_DMA_REGION22_HIGH_REG	Region22 end address configuration register	0x00BC	R/W
PMS_DMA_REGION23_LOW_REG	Region23 start address configuration register	0x00C0	R/W
PMS_DMA_REGION23_HIGH_REG	Region23 end address configuration register	0x00C4	R/W
PMS_DMA_REGION24_LOW_REG	Region24 start address configuration register	0x00C8	R/W
PMS_DMA_REGION24_HIGH_REG	Region24 end address configuration register	0x00CC	R/W
PMS_DMA_REGION25_LOW_REG	Region25 start address configuration register	0x00D0	R/W
PMS_DMA_REGION25_HIGH_REG	Region25 end address configuration register	0x00D4	R/W
PMS_DMA_REGION26_LOW_REG	Region26 start address configuration register	0x00D8	R/W
PMS_DMA_REGION26_HIGH_REG	Region26 end address configuration register	0x00DC	R/W
PMS_DMA_REGION27_LOW_REG	Region27 start address configuration register	0x00E0	R/W
PMS_DMA_REGION27_HIGH_REG	Region27 end address configuration register	0x00E4	R/W
PMS_DMA_REGION28_LOW_REG	Region28 start address configuration register	0x00E8	R/W

Name	Description	Address	Access
PMS_DMA_REGION28_HIGH_REG	Region28 end address configuration register	0x00EC	R/W
PMS_DMA_REGION29_LOW_REG	Region29 start address configuration register	0x00F0	R/W
PMS_DMA_REGION29_HIGH_REG	Region29 end address configuration register	0x00F4	R/W
PMS_DMA_REGION30_LOW_REG	Region30 start address configuration register	0x00F8	R/W
PMS_DMA_REGION30_HIGH_REG	Region30 end address configuration register	0x00FC	R/W
PMS_DMA_REGION31_LOW_REG	Region31 start address configuration register	0x0100	R/W
PMS_DMA_REGION31_HIGH_REG	Region31 end address configuration register	0x0104	R/W
DMA Masters Read and Write Permission Control Registers			
PMS_DMA_GDMA_CH0_R_PMS_REG	GDMA ch0 read permission control register	0x0108	R/W
PMS_DMA_GDMA_CH0_W_PMS_REG	GDMA ch0 write permission control register	0x010C	R/W
PMS_DMA_GDMA_CH1_R_PMS_REG	GDMA ch1 read permission control register	0x0110	R/W
PMS_DMA_GDMA_CH1_W_PMS_REG	GDMA ch1 write permission control register	0x0114	R/W
PMS_DMA_GDMA_CH2_R_PMS_REG	GDMA ch2 read permission control register	0x0118	R/W
PMS_DMA_GDMA_CH2_W_PMS_REG	GDMA ch2 write permission control register	0x011C	R/W
PMS_DMA_GDMA_CH3_R_PMS_REG	GDMA ch3 read permission control register	0x0120	R/W
PMS_DMA_GDMA_CH3_W_PMS_REG	GDMA ch3 write permission control register	0x0124	R/W
PMS_DMA_AHB_PDMA_ADC_R_PMS_REG	GDMA-AHB ADC read permission control register	0x0128	R/W
PMS_DMA_AHB_PDMA_ADC_W_PMS_REG	GDMA-AHB ADC write permission control register	0x012C	R/W
PMS_DMA_AHB_PDMA_I2S0_R_PMS_REG	GDMA-AHB I2S0 read permission control register	0x0130	R/W
PMS_DMA_AHB_PDMA_I2S0_W_PMS_REG	GDMA-AHB I2S0 write permission control register	0x0134	R/W
PMS_DMA_AHB_PDMA_I2S1_R_PMS_REG	GDMA-AHB I2S1 read permission control register	0x0138	R/W
PMS_DMA_AHB_PDMA_I2S1_W_PMS_REG	GDMA-AHB I2S1 write permission control register	0x013C	R/W

Name	Description	Address	Access
PMS_DMA_AHB_PDMA_I2S2_R_PMS_REG	GDMA-AHB I2S2 read permission control register	0x0140	R/W
PMS_DMA_AHB_PDMA_I2S2_W_PMS_REG	GDMA-AHB I2S2 write permission control register	0x0144	R/W
PMS_DMA_AHB_PDMA_I3C_MST_R_PMS_REG	GDMA-AHB I3C MST read permission control register	0x0148	R/W
PMS_DMA_AHB_PDMA_I3C_MST_W_PMS_REG	GDMA-AHB I3C MST write permission control register	0x014C	R/W
PMS_DMA_AHB_PDMA_UHCIO_R_PMS_REG	GDMA-AHB UHCI read permission control register	0x0150	R/W
PMS_DMA_AHB_PDMA_UHCIO_W_PMS_REG	GDMA-AHB UHCI write permission control register	0x0154	R/W
PMS_DMA_AHB_PDMA_RMT_R_PMS_REG	GDMA-AHB RMT read permission control register	0x0158	R/W
PMS_DMA_AHB_PDMA_RMT_W_PMS_REG	GDMA-AHB RMT write permission control register	0x0170	R/W
PMS_DMA_AXI_PDMA_LCDCAM_R_PMS_REG	GDMA-AXI LCD_CAM read permission control register	0x0174	R/W
PMS_DMA_AXI_PDMA_LCDCAM_W_PMS_REG	GDMA-AXI LCD_CAM write permission control register	0x0178	R/W
PMS_DMA_AXI_PDMA_GPSPI2_R_PMS_REG	GDMA-AXI GPSPI2 read permission control register	0x017C	R/W
PMS_DMA_AXI_PDMA_GPSPI2_W_PMS_REG	GDMA-AXI GPSPI2 write permission control register	0x0180	R/W
PMS_DMA_AXI_PDMA_GPSPI3_R_PMS_REG	GDMA-AXI GPSPI3 read permission control register	0x0184	R/W
PMS_DMA_AXI_PDMA_GPSPI3_W_PMS_REG	GDMA-AXI GPSPI3 write permission control register	0x0188	R/W
PMS_DMA_AXI_PDMA_PARLIO_R_PMS_REG	GDMA-AXI PARLIO read permission control register	0x018C	R/W
PMS_DMA_AXI_PDMA_PARLIO_W_PMS_REG	GDMA-AXI PARLIO write permission control register	0x0190	R/W
PMS_DMA_AXI_PDMA_AES_R_PMS_REG	GDMA-AXI AES read permission control register	0x0194	R/W
PMS_DMA_AXI_PDMA_AES_W_PMS_REG	GDMA-AXI AES write permission control register	0x0198	R/W
PMS_DMA_AXI_PDMA_SHA_R_PMS_REG	GDMA-AXI SHA read permission control register	0x019C	R/W
PMS_DMA_AXI_PDMA_SHA_W_PMS_REG	GDMA-AXI SHA write permission control register	0x01A0	R/W
PMS_DMA_DMA2D_JPEG_PMS_R_REG	2D-DMA JPEG read permission control register	0x01A4	R/W

Name	Description	Address	Access
PMS_DMA_DMA2D_JPEG_PMS_W_REG	2D-DMA JPEG write permission control register	0x01A8	R/W
PMS_DMA_USB_PMS_R_REG	High-speed USB 2.0 OTG read permission control register	0x01AC	R/W
PMS_DMA_USB_PMS_W_REG	High-speed USB 2.0 OTG write permission control register	0x01B0	R/W
PMS_DMA_GMAC_PMS_R_REG	EMAC read permission control register	0x01B4	R/W
PMS_DMA_GMAC_PMS_W_REG	EMAC write permission control register	0x01B8	R/W
PMS_DMA_SDMMC_PMS_R_REG	SDMMC read permission control register	0x01BC	R/W
PMS_DMA_SDMMC_PMS_W_REG	SDMMC write permission control register	0x01C0	R/W
PMS_DMA_USBOTG11_PMS_R_REG	Full-speed USB 2.0 OTG read permission control register	0x01C4	R/W
PMS_DMA_USBOTG11_PMS_W_REG	Full-speed USB 2.0 OTG write permission control register	0x01C8	R/W
PMS_DMA_TRACE0_PMS_R_REG	TRACE0 read permission control register	0x01CC	R/W
PMS_DMA_TRACE0_PMS_W_REG	TRACE0 write permission control register	0x01D0	R/W
PMS_DMA_TRACE1_PMS_R_REG	TRACE1 read permission control register	0x01D4	R/W
PMS_DMA_TRACE1_PMS_W_REG	TRACE1 write permission control register	0x01D8	R/W
PMS_DMA_L2MEM_MON_PMS_W_REG	L2MEM Monitor write permission control register	0x01E0	R/W
PMS_DMA_SPM_MON_PMS_W_REG	SPM Monitor write permission control register	0x01E8	R/W
PMS_DMA_H264_PMS_R_REG	H264 DMA read permission control register	0x01FC	R/W
PMS_DMA_H264_PMS_W_REG	H264 DMA write permission control register	0x0200	R/W
PMS_DMA_DMA2D_PPA_PMS_R_REG	2D-DMA PPA read permission control register	0x0204	R/W
PMS_DMA_DMA2D_PPA_PMS_W_REG	2D-DMA PPA write permission control register	0x0208	R/W
PMS_DMA_DMA2D_DUMMY_PMS_R_REG	2D-DMA Dummy read permission control register	0x020C	R/W
PMS_DMA_DMA2D_DUMMY_PMS_W_REG	2D-DMA Dummy write permission control register	0x0210	R/W
PMS_DMA_AHB_PDMA_DUMMY_R_PMS_REG	GDMA-AHB Dummy read permission control register	0x0214	R/W

Name	Description	Address	Access
PMS_DMA_AHB_PDMA_DUMMY_W_PMS_REG	GDMA-AHB Dummy write permission control register	0x0218	R/W
PMS_DMA_AXI_PDMA_DUMMY_R_PMS_REG	GDMA-AXI Dummy read permission control register	0x021C	R/W
PMS_DMA_AXI_PDMA_DUMMY_W_PMS_REG	GDMA-AXI Dummy write permission control register	0x0220	R/W

19.6.2 HP_PERI_PMS_REG

The addresses in this section are relative to the HP_PERI_PMS base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section [Access Types for Registers](#).

Name	Description	Address	Access
Version Control Registers			
PMS_HP_PERI_PMS_DATE_REG	Version control register	0x0000	R/W
Clock Gating Registers			
PMS_HP_PERI_PMS_CLK_EN_REG	Clock gating register	0x0004	R/W
HP CPU Permission Control Registers			
PMS_CORE0_MM_HP_PERI_PMS_REG0_REG	Permission control register0 for HP CPU0 in machine mode	0x0008	R/W
PMS_CORE0_MM_HP_PERI_PMS_REG1_REG	Permission control register1 for HP CPU0 in machine mode	0x000C	R/W
PMS_CORE0_MM_HP_PERI_PMS_REG2_REG	Permission control register2 for HP CPU0 in machine mode	0x0010	R/W
PMS_CORE0_MM_HP_PERI_PMS_REG3_REG	Permission control register3 for HP CPU0 in machine mode	0x0014	R/W
PMS_CORE0_UM_HP_PERI_PMS_REG0_REG	Permission control register0 for HP CPU0 in user mode	0x0018	R/W
PMS_CORE0_UM_HP_PERI_PMS_REG1_REG	Permission control register1 for HP CPU0 in user mode	0x001C	R/W
PMS_CORE0_UM_HP_PERI_PMS_REG2_REG	Permission control register2 for HP CPU0 in user mode	0x0020	R/W
PMS_CORE0_UM_HP_PERI_PMS_REG3_REG	Permission control register3 for HP CPU0 in user mode	0x0024	R/W
PMS_CORE1_MM_HP_PERI_PMS_REG0_REG	Permission control register0 for HP CPU1 in machine mode	0x0028	R/W
PMS_CORE1_MM_HP_PERI_PMS_REG1_REG	Permission control register1 for HP CPU1 in machine mode	0x002C	R/W
PMS_CORE1_MM_HP_PERI_PMS_REG2_REG	Permission control register2 for HP CPU1 in machine mode	0x0030	R/W

Name	Description	Address	Access
PMS_CORE1_MM_HP_PERI_PMS_REG3_REG	Permission control register3 for HP CPU1 in machine mode	0x0034	R/W
PMS_CORE1_UM_HP_PERI_PMS_REG0_REG	Permission control register0 for HP CPU1 in user mode	0x0038	R/W
PMS_CORE1_UM_HP_PERI_PMS_REG1_REG	Permission control register1 for HP CPU1 in user mode	0x003C	R/W
PMS_CORE1_UM_HP_PERI_PMS_REG2_REG	Permission control register2 for HP CPU1 in user mode	0x0040	R/W
PMS_CORE1_UM_HP_PERI_PMS_REG3_REG	Permission control register3 for HP CPU1 in user mode	0x0044	R/W

19.6.3 LP2HP_PERI_PMS_REG

The addresses in this section are relative to the LP2HP_PERI_PMS base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section [Access Types for Registers](#).

Name	Description	Address	Access
Version Control Registers			
PMS_LP2HP_PERI_PMS_DATE_REG	Version control register	0x0000	R/W
Clock Gating Registers			
PMS_LP2HP_PERI_PMS_CLK_EN_REG	Clock gating register	0x0004	R/W
LP CPU Permission Control Registers			
PMS_LP_MM_PMS_REG0_REG	Permission control register0 for the LP CPU in machine mode	0x0008	R/W
PMS_LP_MM_PMS_REG1_REG	Permission control register1 for the LP CPU in machine mode	0x0030	R/W
PMS_LP_MM_PMS_REG2_REG	Permission control register2 for the LP CPU in machine mode	0x00A4	R/W
PMS_LP_MM_PMS_REG3_REG	Permission control register3 for the LP CPU in machine mode	0x011C	R/W

19.6.4 LP_PERI_PMS_REG

The addresses in this section are relative to the LP_PERI_PMS base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section [Access Types for Registers](#).

Name	Description	Address	Access
Version Control Registers			

Name	Description	Address	Access
PMS_LP_PERI_PMS_DATE_REG	Version control register	0x0000	R/W
Clock Gating Registers			
PMS_LP_PERI_PMS_CLK_EN_REG	Clock gating register	0x0004	R/W
LP CPU Permission Control Registers			
PMS_LP_MM_LP_PERI_PMS_REGO_REG	Permission control register0 for LP CPU in machine mode	0x0008	R/W
Configurable Address Range Configuration Registers			
PMS_PERI_REGION0_LOW_REG	Region0 start address configuration register	0x000C	R/W
PMS_PERI_REGION0_HIGH_REG	Region0 end address configuration register	0x0010	R/W
PMS_PERI_REGION1_LOW_REG	Region1 start address configuration register	0x0014	R/W
PMS_PERI_REGION_PMS_REG	Permission control of access to address region 0-1	0x001C	R/W
PMS_PERI_REGION_2_TO_7_PMS_REG	Permission control of access to address region 2-7	0x0020	R/W
PMS_PERI_REGION2_LOW_REG	Address region 2 start address configuration register	0x0024	R/W
PMS_PERI_REGION2_HIGH_REG	Address region 2 end address configuration register	0x0028	R/W
PMS_PERI_REGION3_LOW_REG	Address region 3 start address configuration register	0x002C	R/W
PMS_PERI_REGION3_HIGH_REG	Address region 3 end address configuration register	0x0030	R/W
PMS_PERI_REGION4_LOW_REG	Address region 4 start address configuration register	0x0034	R/W
PMS_PERI_REGION4_HIGH_REG	Address region 4 end address configuration register	0x0038	R/W
PMS_PERI_REGION5_LOW_REG	Address region 5 start address configuration register	0x003C	R/W
PMS_PERI_REGION5_HIGH_REG	Address region 5 end address configuration register	0x0040	R/W
PMS_PERI_REGION6_LOW_REG	Address region 6 start address configuration register	0x0044	R/W
PMS_PERI_REGION6_HIGH_REG	Address region 6 end address configuration register	0x0048	R/W
PMS_PERI_REGION7_LOW_REG	Address region 7 start address configuration register	0x004C	R/W
PMS_PERI_REGION7_HIGH_REG	Address region 7 end address configuration register	0x0050	R/W
PMS_PERI_REGION1_HIGH_REG	Region1 end address configuration register	0x0018	R/W

19.6.5 HP2LP_PERI_PMS_REG

The addresses in this section are relative to the HP2LP_PERI_PMS base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Version Control Registers			
PMS_HP2LP_PERI_PMS_DATE_REG	Version control register	0x0000	R/W
Clock Gating Registers			
PMS_HP2LP_PERI_PMS_CLK_EN_REG	Clock gating register	0x0004	R/W
HP CPU Permission Control Registers			
PMS_HP_CORE0_MM_PMS_REGO_REG	Permission control register0 for HP CPU0 in machine mode	0x0008	R/W
PMS_HP_CORE0_UM_PMS_REGO_REG	Permission control register0 for HP CPU0 in user mode	0x000C	R/W
PMS_HP_CORE1_MM_PMS_REGO_REG	Permission control register0 for HP CPU1 in machine mode	0x0010	R/W
PMS_HP_CORE1_UM_PMS_REGO_REG	Permission control register0 for HP CPU1 in user mode	0x0014	R/W

19.7 Registers

19.7.1 HP_DMA_PMS_REG

The addresses in this section are relative to the HP_DMA_PMS base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 19.1. PMS_DMA_DATE_REG (0x0000)

PMS_DMA_DATE	
31	0
0x20230314	
Reset	

PMS_DMA_DATE Version control register. (R/W)

Register 19.2. PMS_DMA_CLK_EN_REG (0x0004)

(reserved)																												PMS_DMA_CLK_EN																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
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PMS_DMA_CLK_EN Configures whether to keep the clock always on.

0: Enable automatic clock gating.

1: Keep the clock always on.

(R/W)

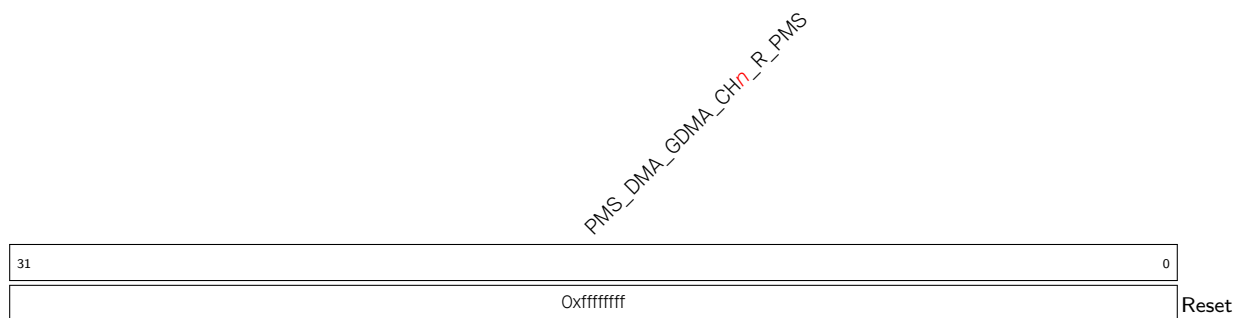
Register 19.3. PMS_DMA_REGION_{*n*}_LOW_REG (*n*: 0-31) (0x0008+0x8n*)**

PMS_DMA_REGION _{<i>n</i>} _LOW																(reserved)																															
31												12				11												0																			
0x000												0				0				0				0				0				0				0				0				Reset			

PMS_DMA_REGION_{*n*}_LOW Configures the high 20 bits of the start address for region *n*. (R/W)

Register 19.4. PMS_DMA_REGION n _HIGH_REG (n : 0-31) (0x000C+0x8* n)

PMS_DMA_REGION n _HIGH Configures the high 20 bits of the end address for region n . (R/W)

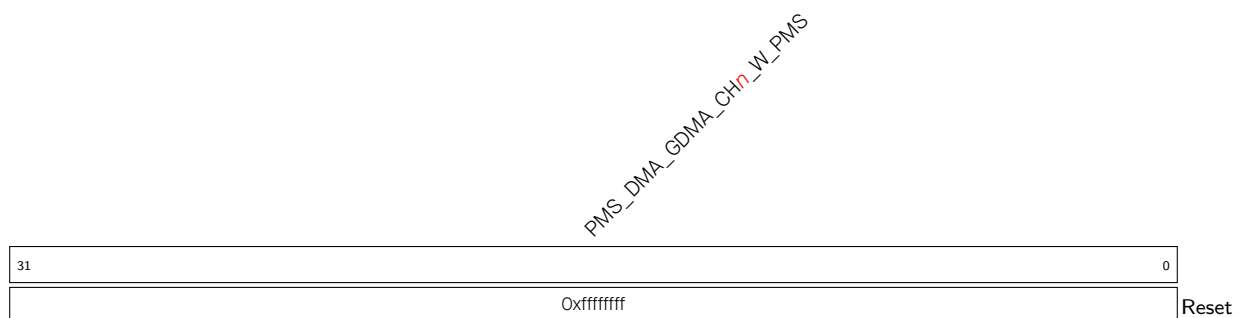
Register 19.5. PMS_DMA_GDMA_CH n _R_PMS_REG (n : 0-3) (0x0108+0x8* n)

PMS_DMA_GDMA_CH n _R_PMS Configures the permission for GDMA ch n to read 32 address regions. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

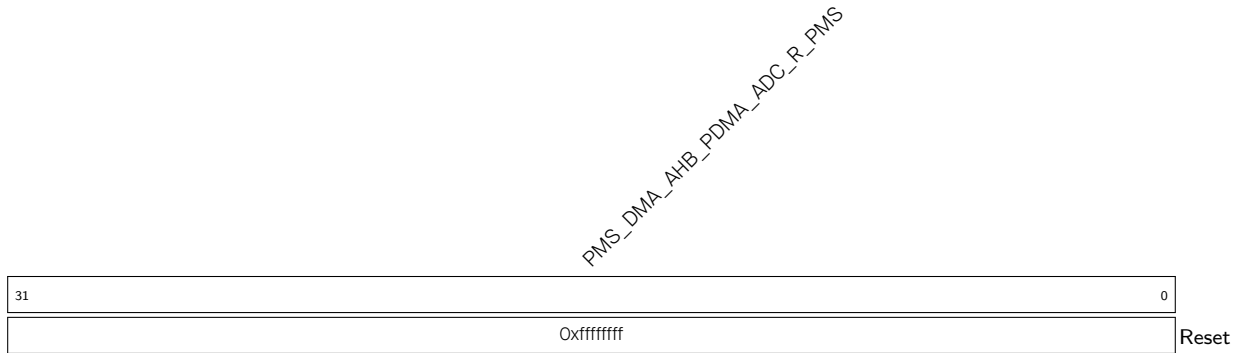
Register 19.6. PMS_DMA_GDMA_CH n _W_PMS_REG (n : 0-3) (0x010C+0x8* n)

PMS_DMA_GDMA_CH n _W_PMS Configures the permission for GDMA ch n to write 32 address regions. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

(R/W)

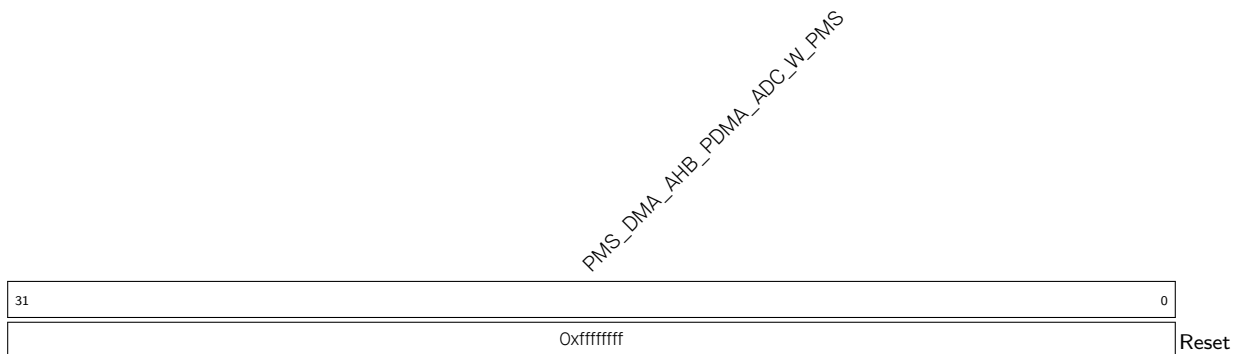
Register 19.7. PMS_DMA_AHB_PDMA_ADC_R_PMS_REG (0x0128)

PMS_DMA_AHB_PDMA_ADC_R_PMS Configures GDMA-AHB permission to read 32 address ranges requested by ADC. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

Register 19.8. PMS_DMA_AHB_PDMA_ADC_W_PMS_REG (0x012C)

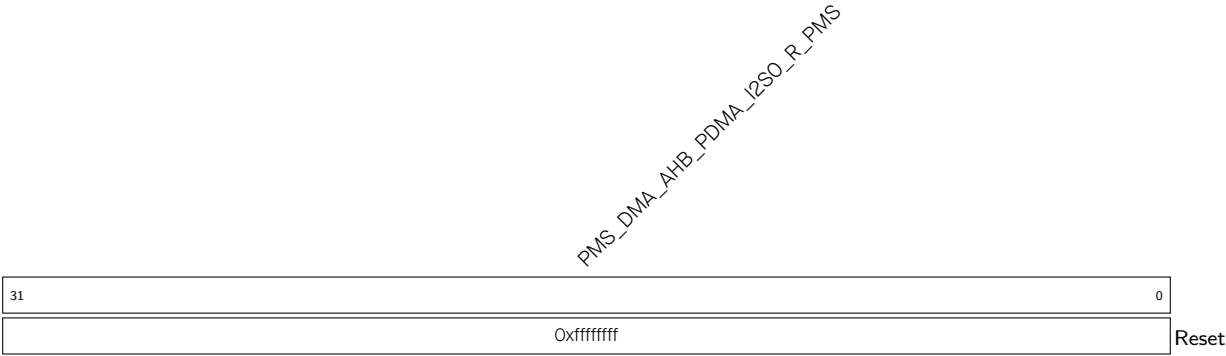
PMS_DMA_AHB_PDMA_ADC_W_PMS Configures GDMA-AHB permission to write 32 address ranges requested by ADC. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

(R/W)

Register 19.9. PMS_DMA_AHB_PDMA_I2SO_R_PMS_REG (0x0130)



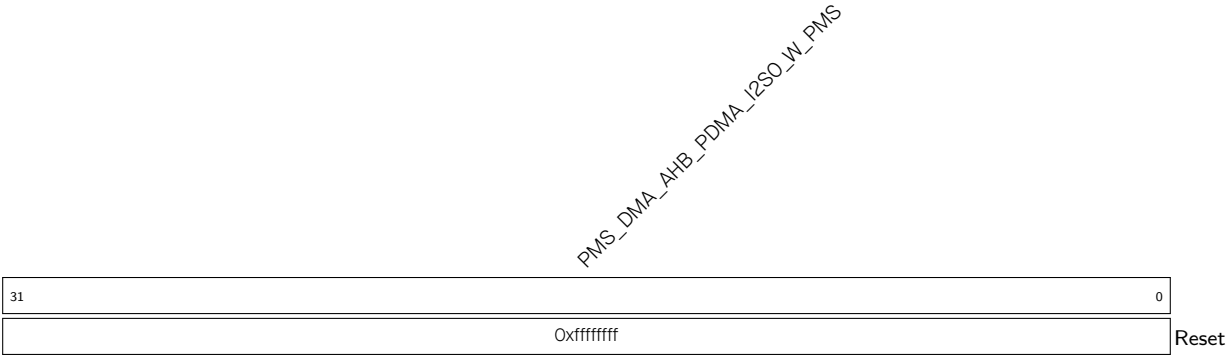
PMS_DMA_AHB_PDMA_I2SO_R_PMS Configures GDMA-AHB permission to read 32 address ranges requested by I2SO. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

Register 19.10. PMS_DMA_AHB_PDMA_I2SO_W_PMS_REG (0x0134)

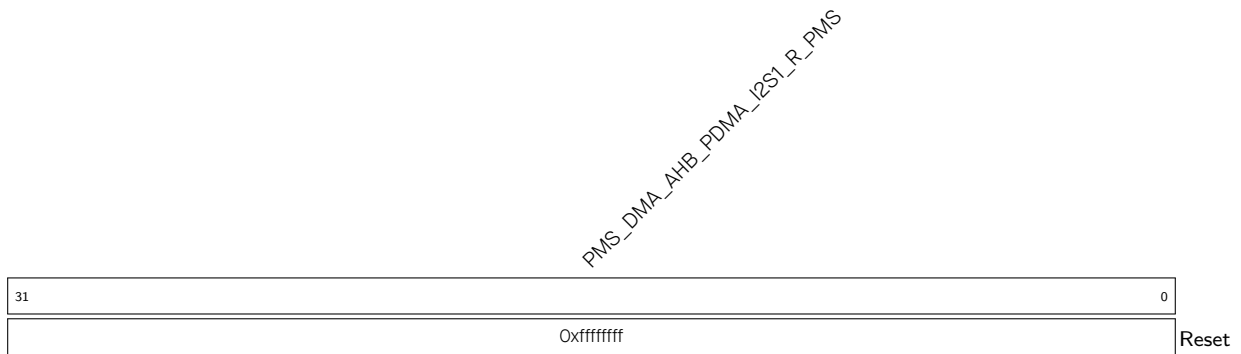


PMS_DMA_AHB_PDMA_I2SO_W_PMS Configures GDMA-AHB permission to write 32 address ranges requested by I2SO. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

(R/W)

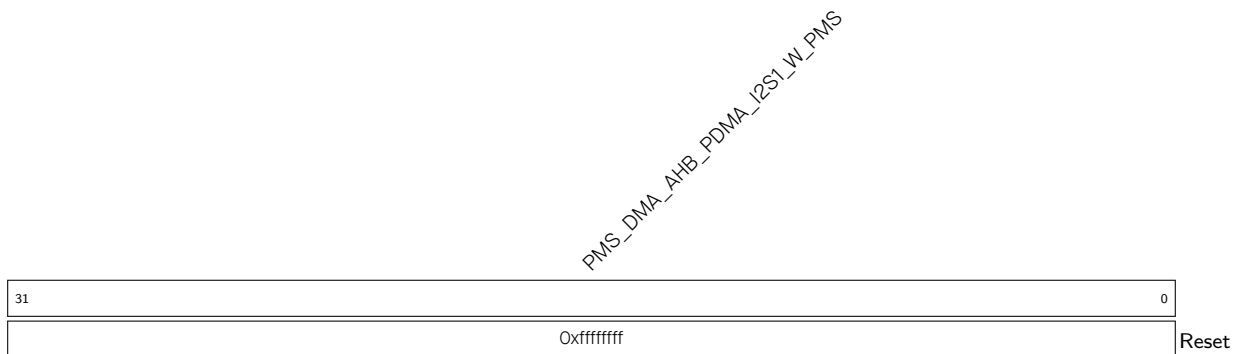
Register 19.11. PMS_DMA_AHB_PDMA_I2S1_R_PMS_REG (0x0138)

PMS_DMA_AHB_PDMA_I2S1_R_PMS Configures GDMA-AHB permission to read 32 address ranges requested by I2S1. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

Register 19.12. PMS_DMA_AHB_PDMA_I2S1_W_PMS_REG (0x013C)

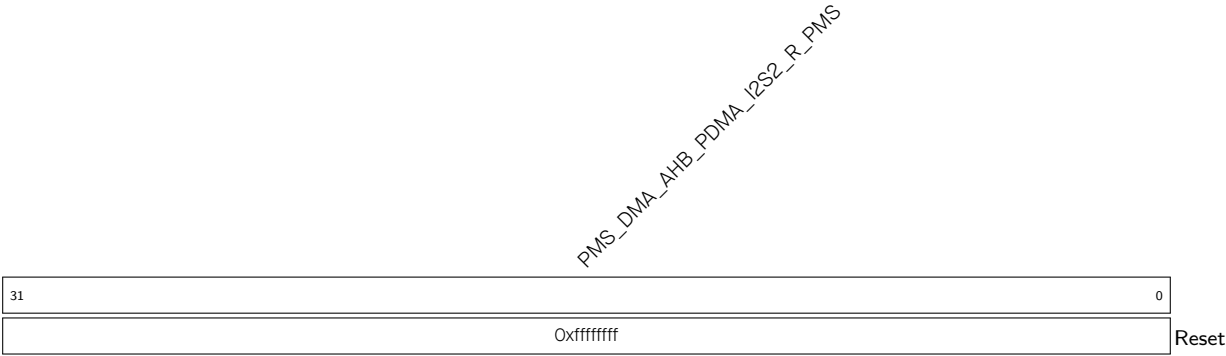
PMS_DMA_AHB_PDMA_I2S1_W_PMS Configures GDMA-AHB permission to write 32 address ranges requested by I2S1. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

(R/W)

Register 19.13. PMS_DMA_AHB_PDMA_I2S2_R_PMS_REG (0x0140)

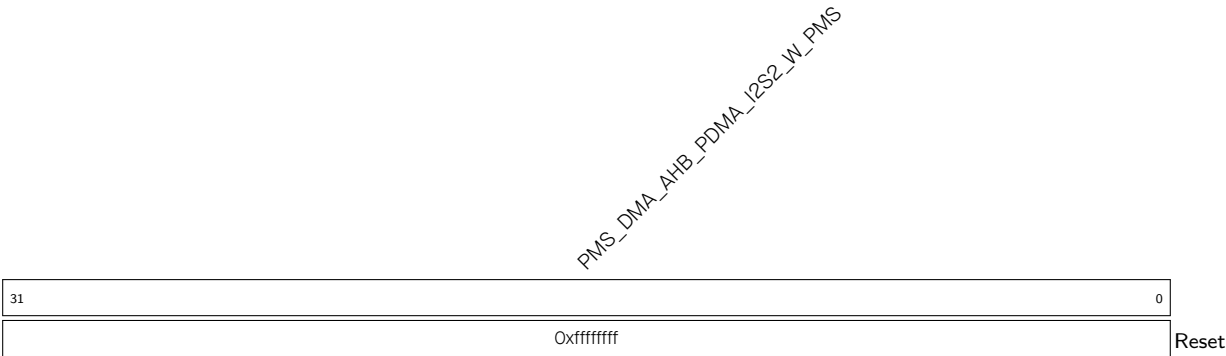


PMS_DMA_AHB_PDMA_I2S2_R_PMS Configures GDMA-AHB permission to read 32 address ranges requested by I2S2. Bit 0 corresponds to region0, and so on.

- 0: Disable read permission.
- 1: Enable read permission.

(R/W)

Register 19.14. PMS_DMA_AHB_PDMA_I2S2_W_PMS_REG (0x0144)

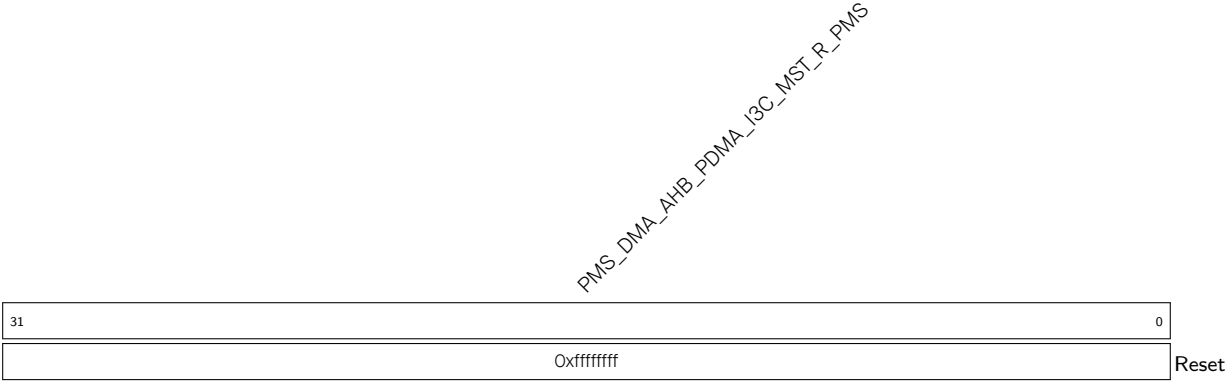


PMS_DMA_AHB_PDMA_I2S2_W_PMS Configures GDMA-AHB permission to write 32 address ranges requested by I2S2. Bit 0 corresponds to region0, and so on.

- 0: Disable write permission.
- 1: Enable write permission.

(R/W)

Register 19.15. PMS_DMA_AHB_PDMA_I3C_MST_R_PMS_REG (0x0148)



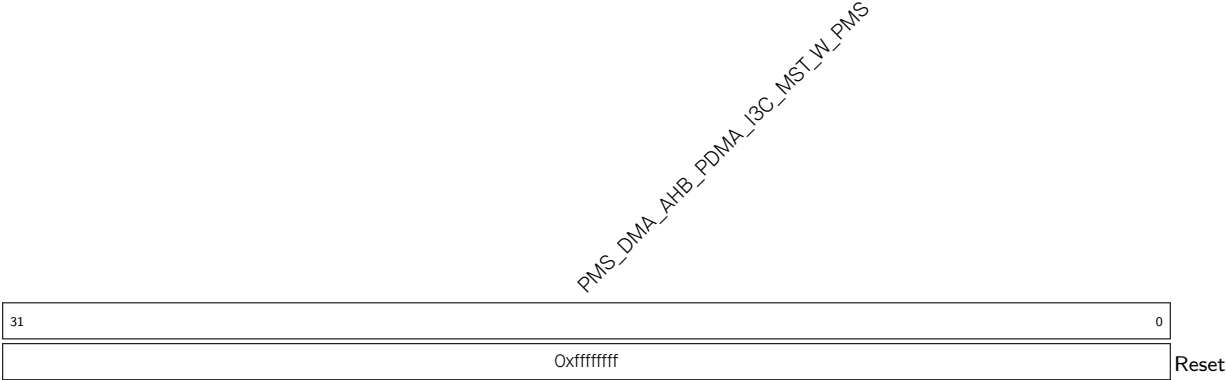
PMS_DMA_AHB_PDMA_I3C_MST_R_PMS Configures GDMA-AHB permission to read 32 address ranges requested by I3C master. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

Register 19.16. PMS_DMA_AHB_PDMA_I3C_MST_W_PMS_REG (0x014C)

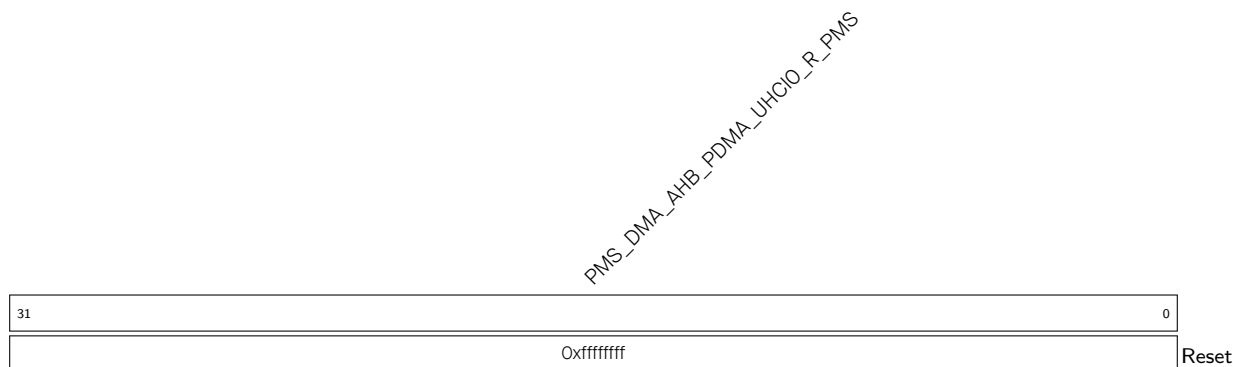


PMS_DMA_AHB_PDMA_I3C_MST_W_PMS Configures GDMA-AHB permission to write 32 address ranges requested by I3C master. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

(R/W)

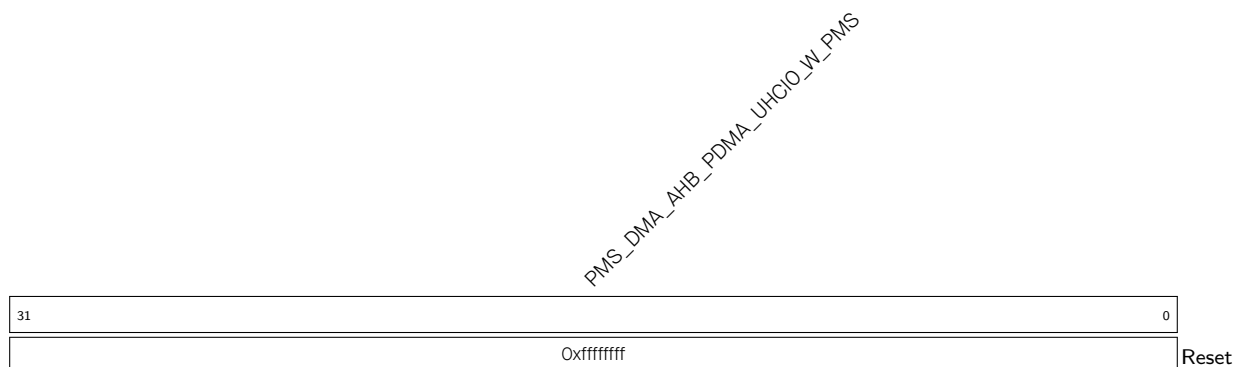
Register 19.17. PMS_DMA_AHB_PDMA_UHCIO_R_PMS_REG (0x0150)

PMS_DMA_AHB_PDMA_UHCIO_R_PMS Configures GDMA-AHB permission to read 32 address ranges requested by UHCI. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

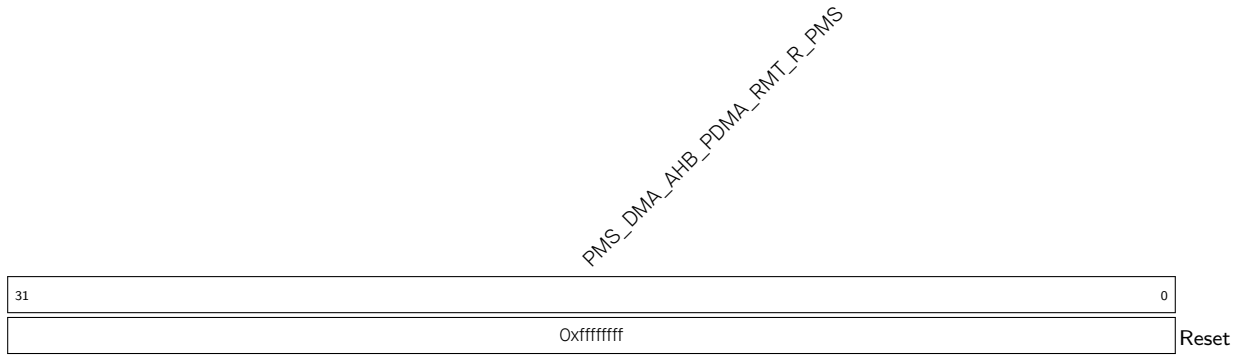
Register 19.18. PMS_DMA_AHB_PDMA_UHCIO_W_PMS_REG (0x0154)

PMS_DMA_AHB_PDMA_UHCIO_W_PMS Configures GDMA-AHB permission to write 32 address ranges requested by UHCI. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

(R/W)

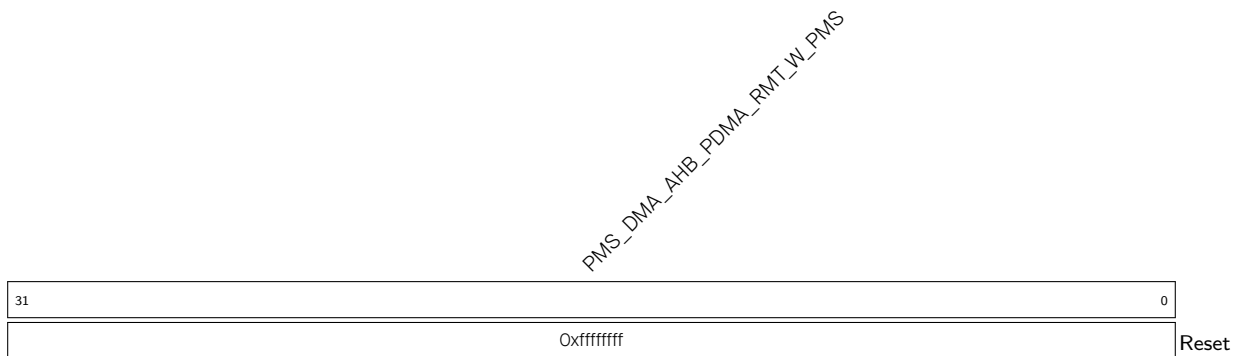
Register 19.19. PMS_DMA_AHB_PDMA_RMT_R_PMS_REG (0x0158)

PMS_DMA_AHB_PDMA_RMT_R_PMS Configures GDMA-AHB permission to read 32 address ranges requested by RMT. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

Register 19.20. PMS_DMA_AHB_PDMA_RMT_W_PMS_REG (0x0170)

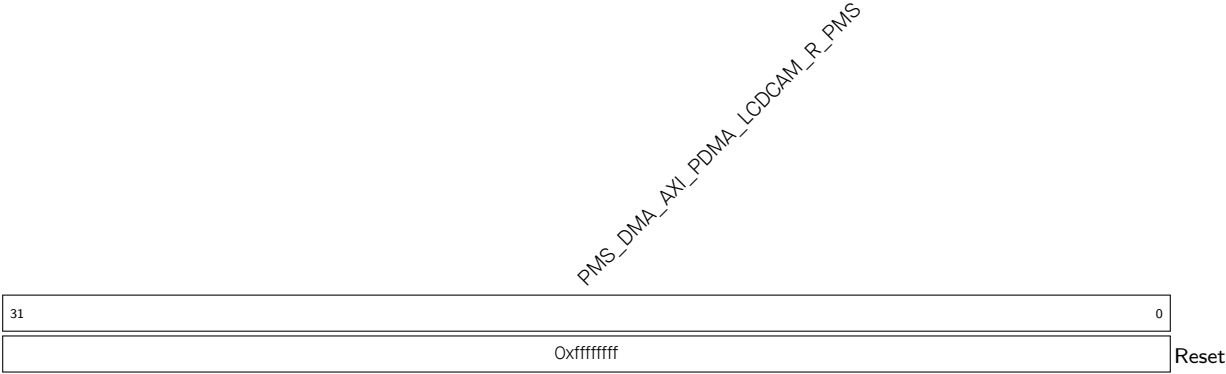
PMS_DMA_AHB_PDMA_RMT_W_PMS Configures GDMA-AHB permission to write 32 address ranges requested by RMT. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

(R/W)

Register 19.21. PMS_DMA_AXI_PDMA_LCDCAM_R_PMS_REG (0x0174)

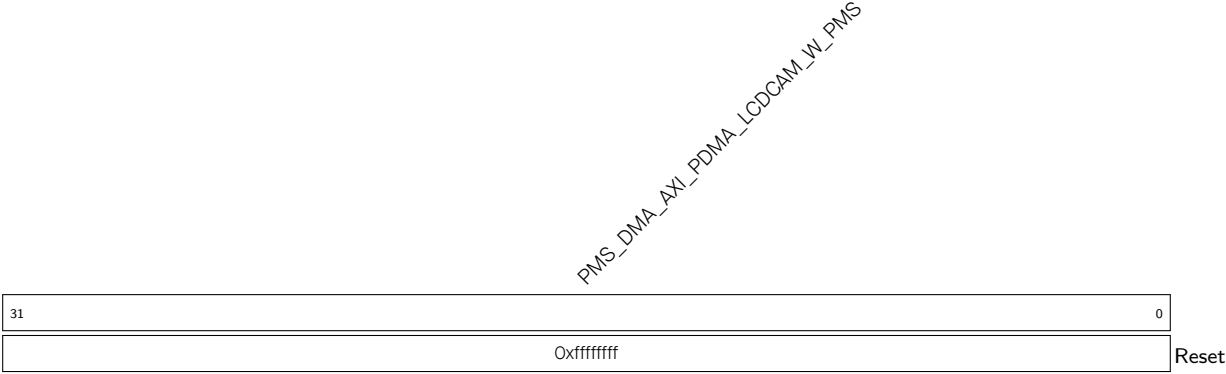


PMS_DMA_AXI_PDMA_LCDCAM_R_PMS Configures GDMA-AXI permission to read 32 address ranges requested by LCD_CAM. Bit 0 corresponds to region0, and so on.

- 0: Disable read permission.
- 1: Enable read permission.

(R/W)

Register 19.22. PMS_DMA_AXI_PDMA_LCDCAM_W_PMS_REG (0x0178)

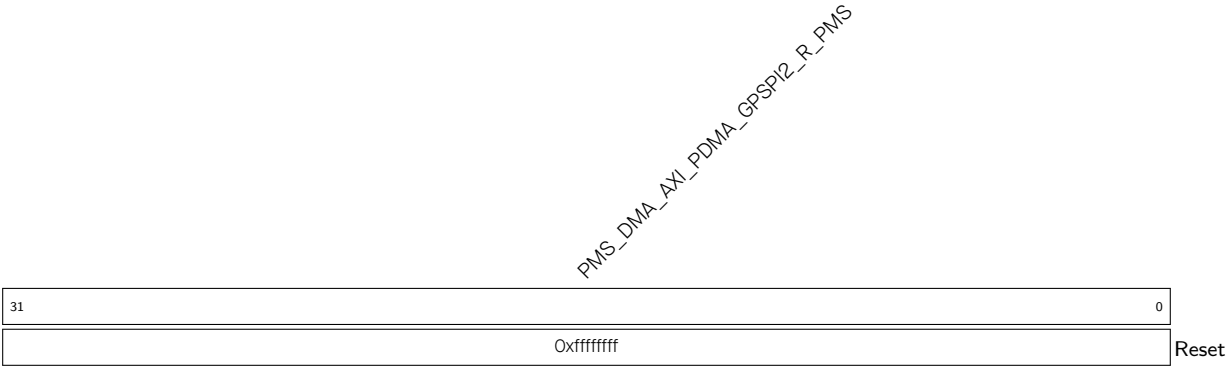


PMS_DMA_AXI_PDMA_LCDCAM_W_PMS Configures GDMA-AXI permission to write 32 address ranges requested by LCD_CAM. Bit 0 corresponds to region0, and so on.

- 0: Disable write permission.
- 1: Enable write permission.

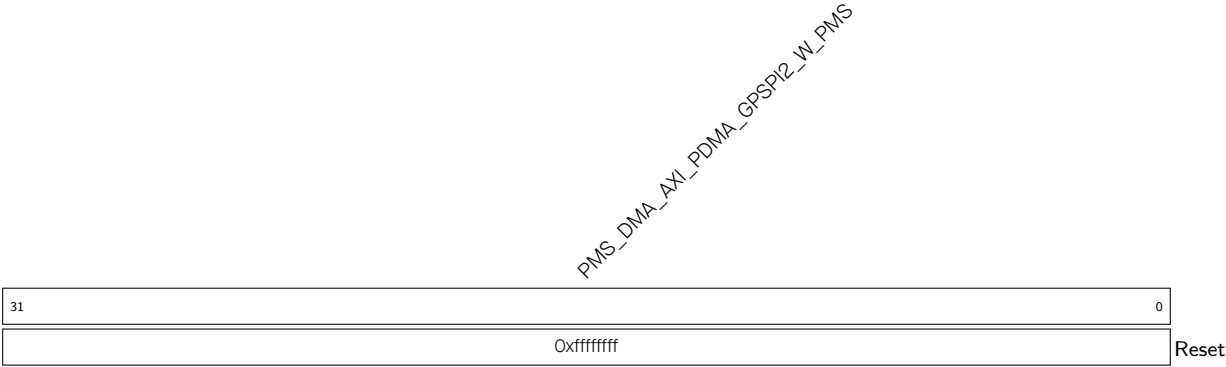
(R/W)

Register 19.23. PMS_DMA_AXI_PDMA_GPSPI2_R_PMS_REG (0x017C)

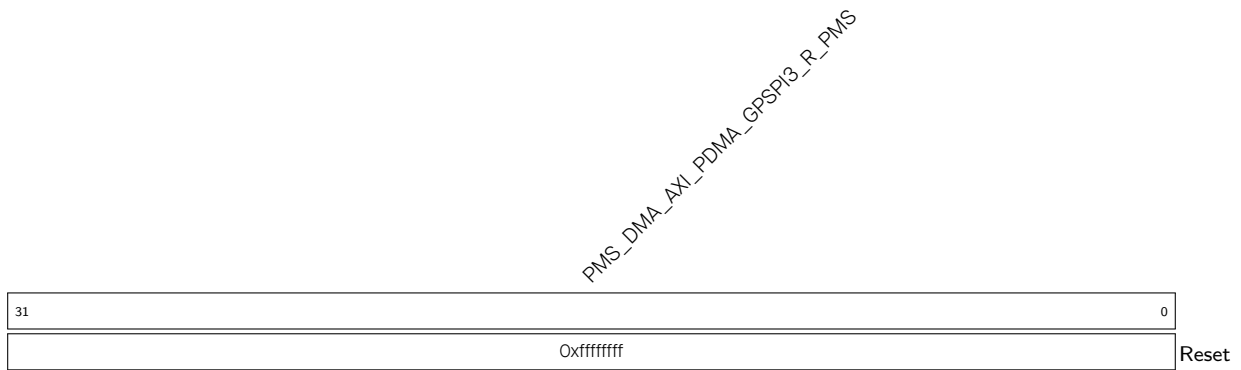


PMS_DMA_AXI_PDMA_GPSPI2_R_PMS Configures GDMA-AXI permission to read 32 address ranges requested by GP-SPI2. Bit 0 corresponds to region0, and so on.
0: Disable read permission.
1: Enable read permission.
(R/W)

Register 19.24. PMS_DMA_AXI_PDMA_GPSPI2_W_PMS_REG (0x0180)



PMS_DMA_AXI_PDMA_GPSPI2_W_PMS Configures GDMA-AXI permission to write 32 address ranges requested by GP-SPI2. Bit 0 corresponds to region0, and so on.
0: Disable write permission.
1: Enable write permission.
(R/W)

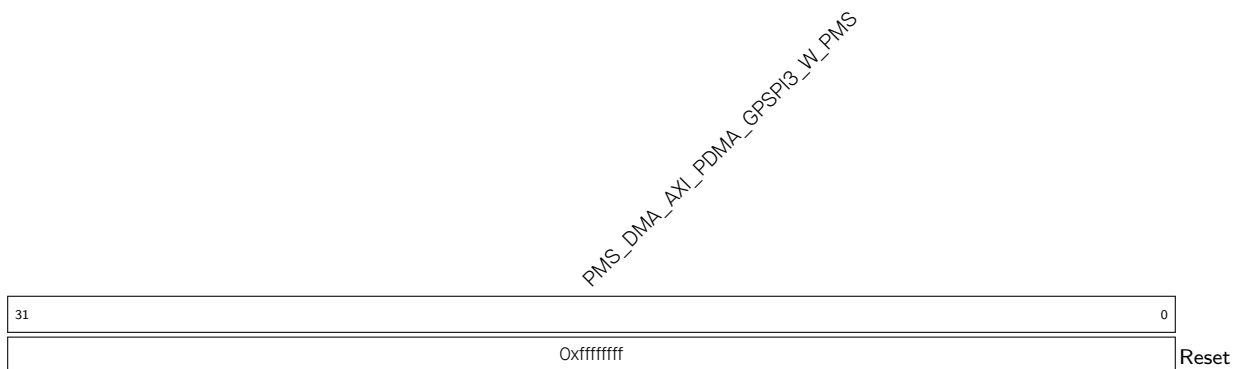
Register 19.25. PMS_DMA_AXI_PDMA_GPSPI3_R_PMS_REG (0x0184)

PMS_DMA_AXI_PDMA_GPSPI3_R_PMS Configures GDMA-AXI permission to read 32 address ranges requested by GP-SPI3. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

Register 19.26. PMS_DMA_AXI_PDMA_GPSPI3_W_PMS_REG (0x0188)

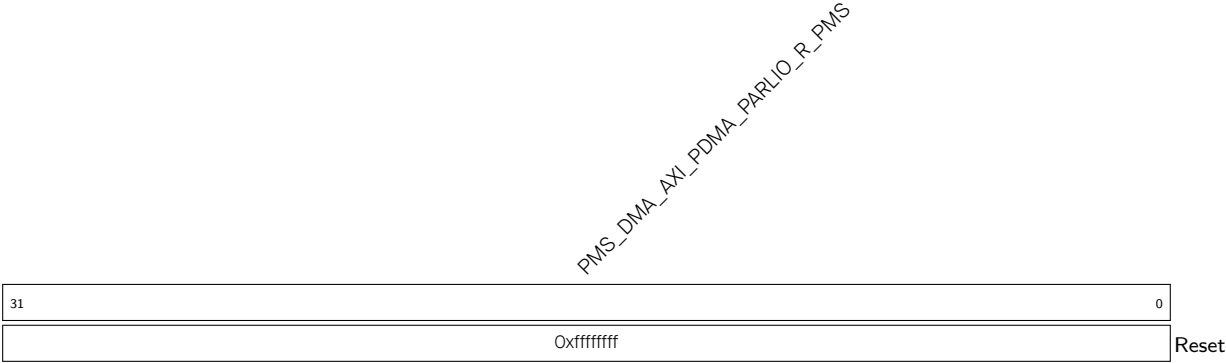
PMS_DMA_AXI_PDMA_GPSPI3_W_PMS Configures GDMA-AXI permission to write 32 address ranges requested by GP-SPI3. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

(R/W)

Register 19.27. PMS_DMA_AXI_PDMA_PARLIO_R_PMS_REG (0x018C)

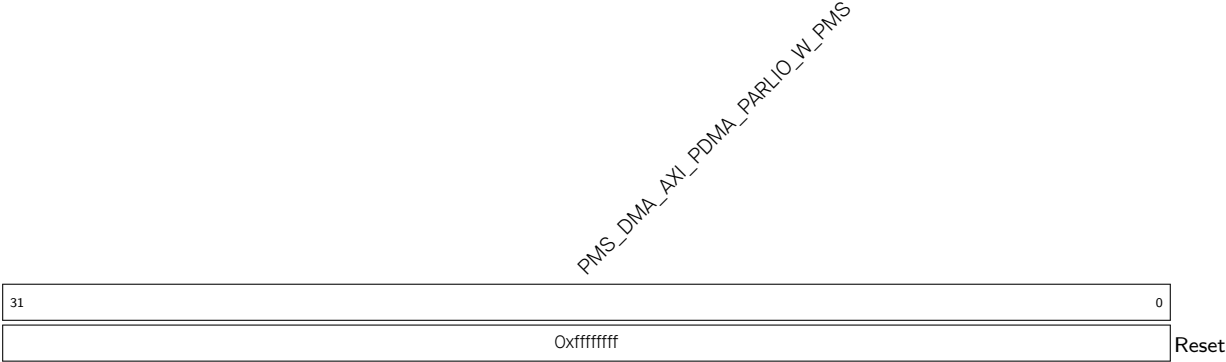


PMS_DMA_AXI_PDMA_PARLIO_R_PMS Configures GDMA-AXI permission to read 32 address ranges requested by PARLIO (Parallel IO Controller). Bit 0 corresponds to region0, and so on.

- 0: Disable read permission.
- 1: Enable read permission.

(R/W)

Register 19.28. PMS_DMA_AXI_PDMA_PARLIO_W_PMS_REG (0x0190)

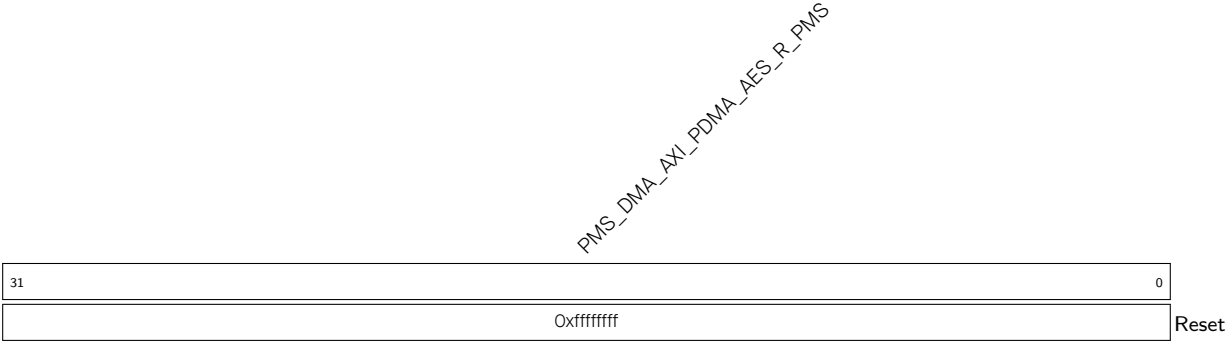


PMS_DMA_AXI_PDMA_PARLIO_W_PMS Configures GDMA-AXI permission to write 32 address ranges requested by PARLIO. Bit 0 corresponds to region0, and so on.

- 0: Disable write permission.
- 1: Enable write permission.

(R/W)

Register 19.29. PMS_DMA_AXI_PDMA_AES_R_PMS_REG (0x0194)

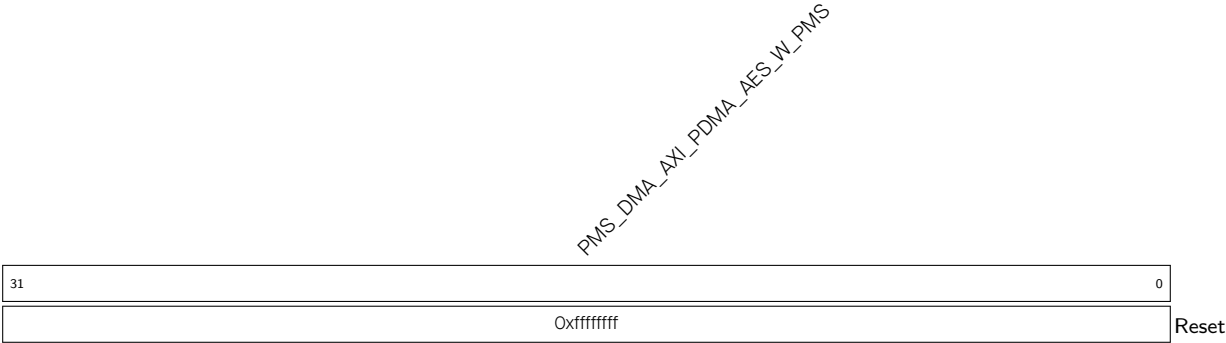


PMS_DMA_AXI_PDMA_AES_R_PMS Configures GDMA-AXI permission to read 32 address ranges requested by AES. Bit 0 corresponds to region0, and so on.

- 0: Disable read permission.
- 1: Enable read permission.

(R/W)

Register 19.30. PMS_DMA_AXI_PDMA_AES_W_PMS_REG (0x0198)

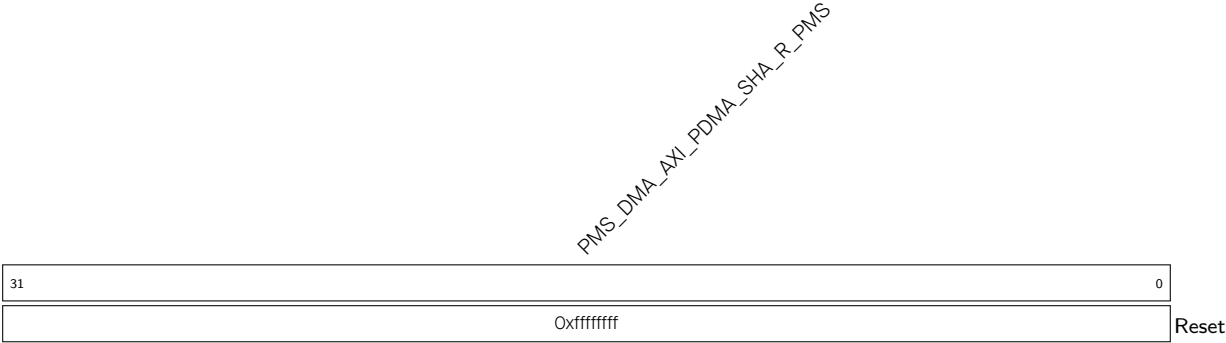


PMS_DMA_AXI_PDMA_AES_W_PMS Configures GDMA-AXI permission to write 32 address ranges requested by AES. Bit 0 corresponds to region0, and so on.

- 0: Disable write permission.
- 1: Enable write permission.

(R/W)

Register 19.31. PMS_DMA_AXI_PDMA_SHA_R_PMS_REG (0x019C)

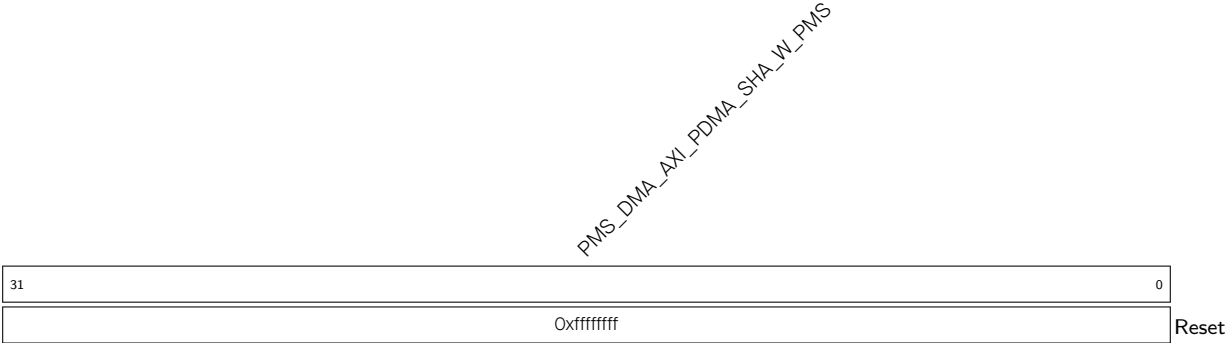


PMS_DMA_AXI_PDMA_SHA_R_PMS Configures GDMA-AXI permission to read 32 address ranges requested by SHA. Bit 0 corresponds to region0, and so on.

- 0: Disable read permission.
- 1: Enable read permission.

(R/W)

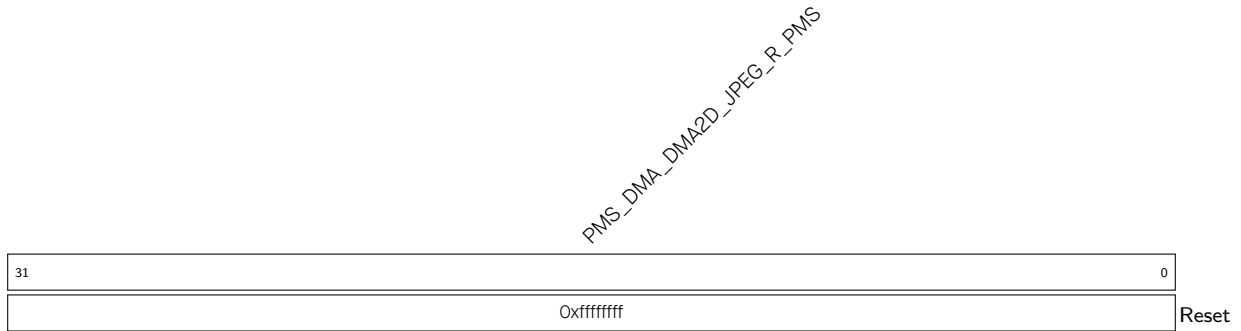
Register 19.32. PMS_DMA_AXI_PDMA_SHA_W_PMS_REG (0x01A0)



PMS_DMA_AXI_PDMA_SHA_W_PMS Configures GDMA-AXI permission to write 32 address ranges requested by SHA. Bit 0 corresponds to region0, and so on.

- 0: Disable write permission.
- 1: Enable write permission.

(R/W)

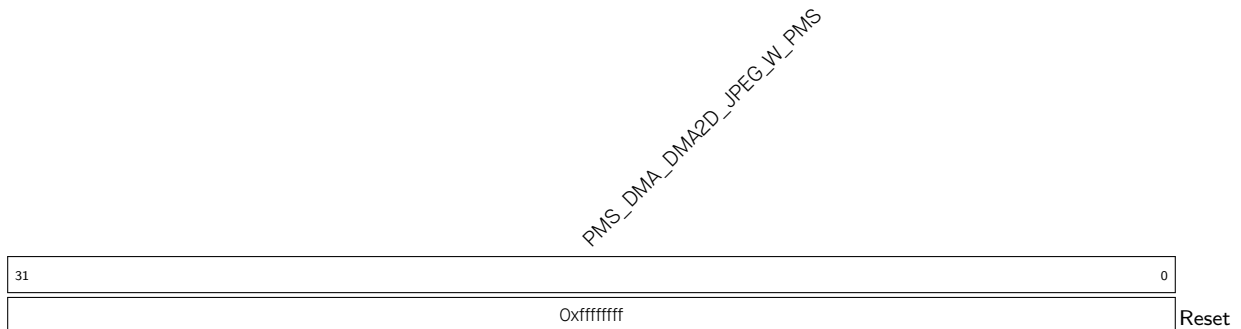
Register 19.33. PMS_DMA_DMA2D_JPEG_PMS_R_REG (0x01A4)

PMS_DMA_DMA2D_JPEG_R_PMS Configures 2D-DMA permission to read 32 address ranges requested by JPEG. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

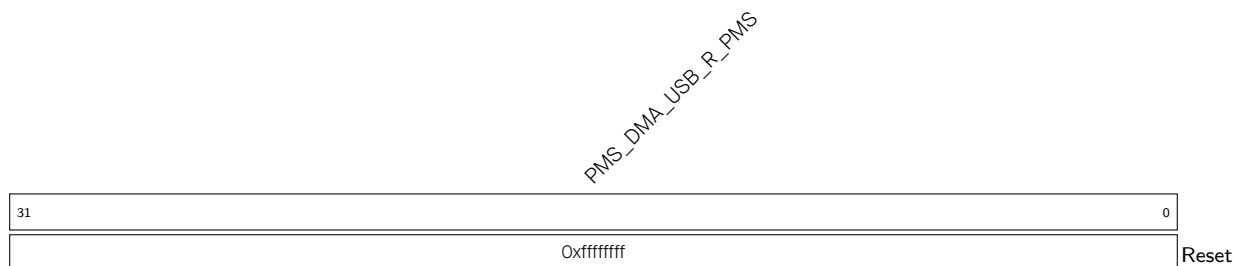
Register 19.34. PMS_DMA_DMA2D_JPEG_PMS_W_REG (0x01A8)

PMS_DMA_DMA2D_JPEG_W_PMS Configures 2D-DMA permission to write 32 address ranges requested by JPEG. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

(R/W)

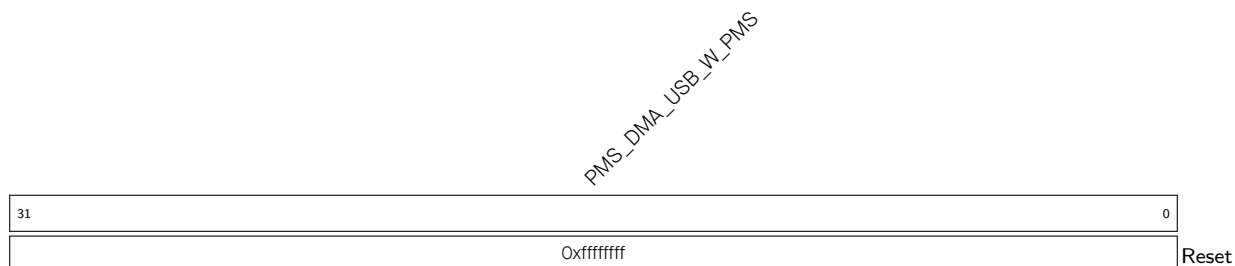
Register 19.35. PMS_DMA_USB_PMS_R_REG (0x01AC)

PMS_DMA_USB_R_PMS Configures read permission for high-speed USB 2.0 OTG to access 32 address ranges. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

Register 19.36. PMS_DMA_USB_PMS_W_REG (0x01B0)

PMS_DMA_USB_W_PMS Configures write permission for high-speed USB 2.0 OTG to access 32 address ranges. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

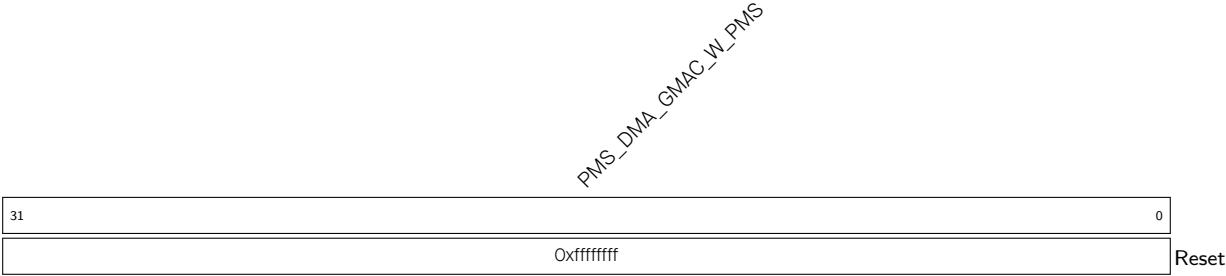
(R/W)

Register 19.37. PMS_DMA_GMAC_PMS_R_REG (0x01B4)



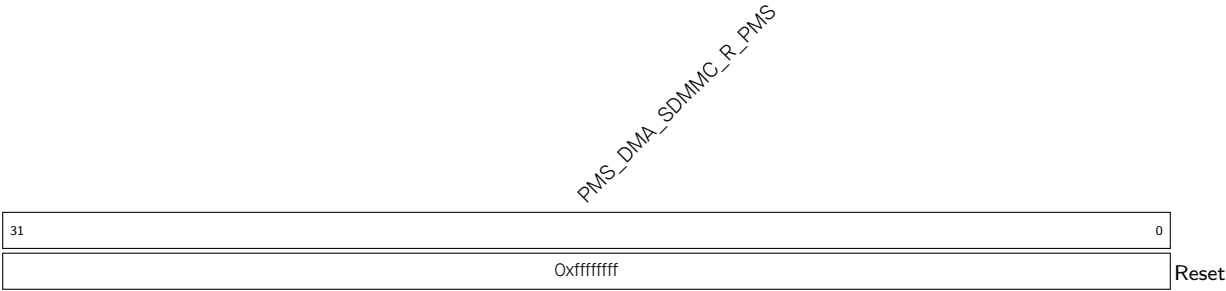
PMS_DMA_GMAC_R_PMS Configures read permission for EMAC to access 32 address ranges. Bit 0 corresponds to region0, and so on.
0: Disable read permission.
1: Enable read permission.
(R/W)

Register 19.38. PMS_DMA_GMAC_PMS_W_REG (0x01B8)



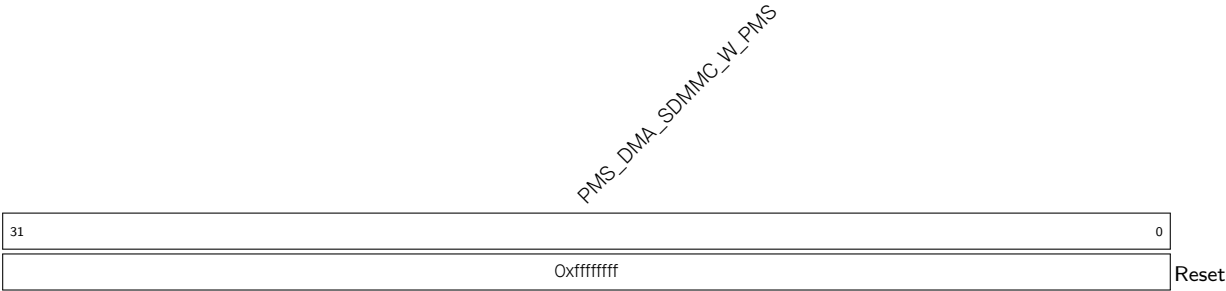
PMS_DMA_GMAC_W_PMS Configures write permission for EMAC to access 32 address ranges. Bit 0 corresponds to region0, and so on.
0: Disable write permission.
1: Enable write permission.
(R/W)

Register 19.39. PMS_DMA_SDMMC_PMS_R_REG (0x01BC)

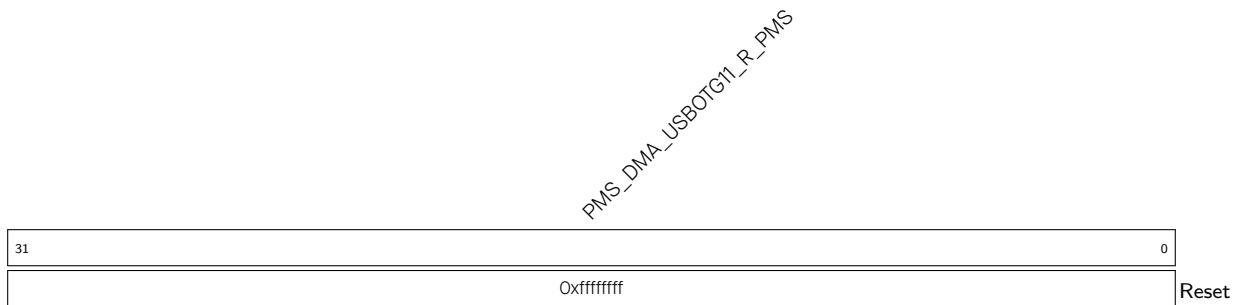


PMS_DMA_SDMMC_R_PMS Configures read permission for SDMMC to access 32 address ranges. Bit 0 corresponds to region0, and so on.
0: Disable read permission.
1: Enable read permission.
(R/W)

Register 19.40. PMS_DMA_SDMMC_PMS_W_REG (0x01C0)



PMS_DMA_SDMMC_W_PMS Configures write permission for SDMMC to access 32 address ranges. Bit 0 corresponds to region0, and so on.
0: Disable write permission.
1: Enable write permission.
(R/W)

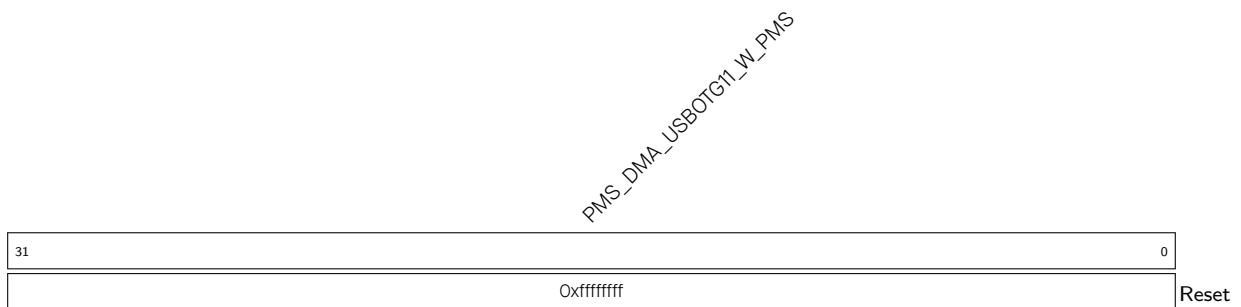
Register 19.41. PMS_DMA_USBOTG11_PMS_R_REG (0x01C4)

PMS_DMA_USBOTG11_R_PMS Configures read permission for full-speed USB 2.0 OTG to access 32 address ranges. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

Register 19.42. PMS_DMA_USBOTG11_PMS_W_REG (0x01C8)

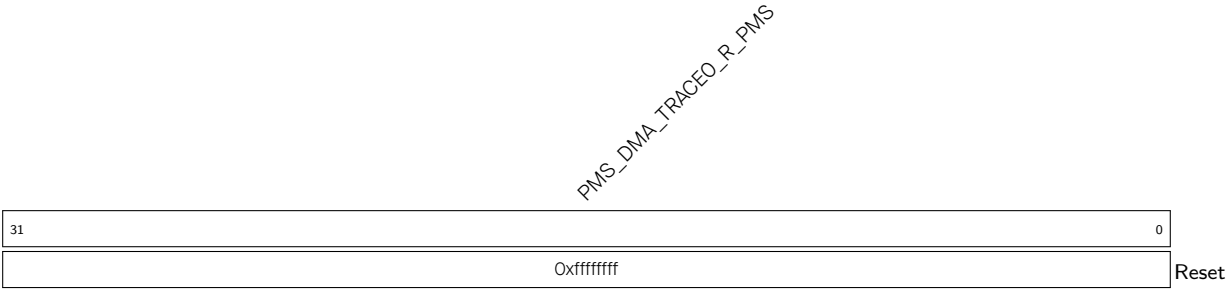
PMS_DMA_USBOTG11_W_PMS Configures write permission for full-speed USB 2.0 OTG to access 32 address ranges. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

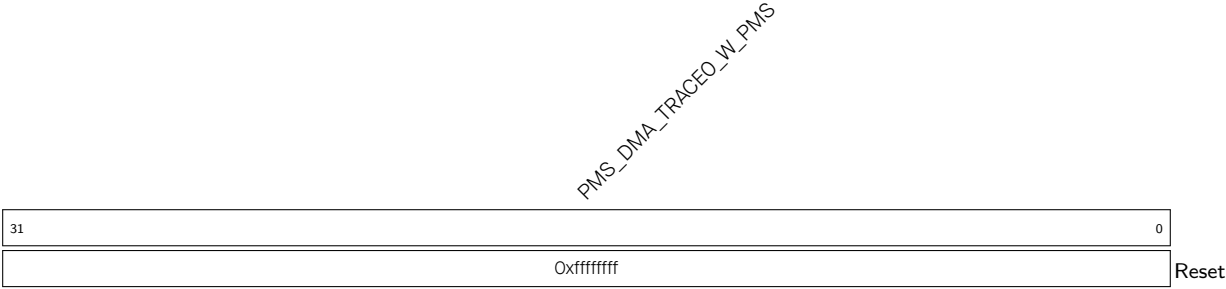
(R/W)

Register 19.43. PMS_DMA_TRACE0_PMS_R_REG (0x01CC)



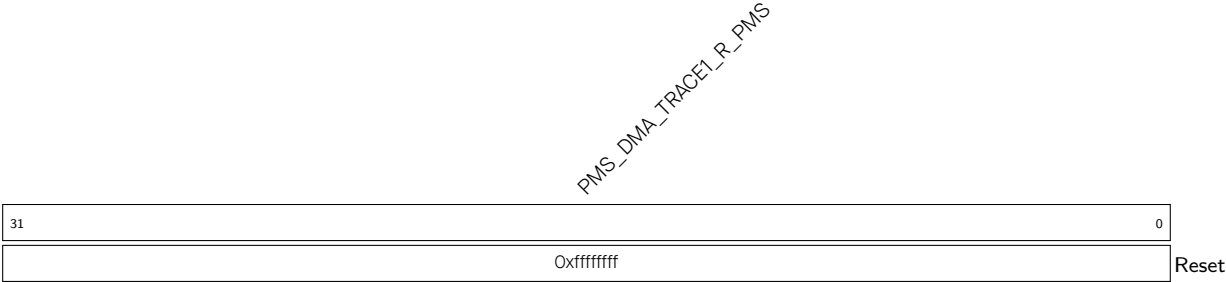
PMS_DMA_TRACE0_R_PMS Configures read permission for TRACE0 to access 32 address ranges.
Bit 0 corresponds to region0, and so on.
0: Disable read permission.
1: Enable read permission.
(R/W)

Register 19.44. PMS_DMA_TRACE0_PMS_W_REG (0x01D0)



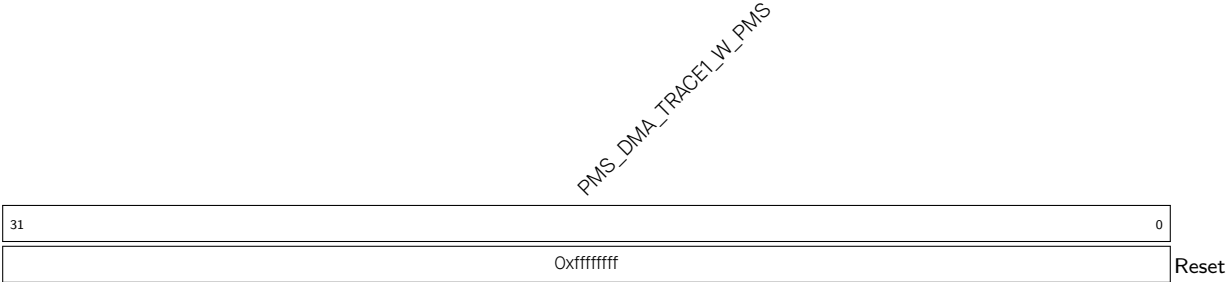
PMS_DMA_TRACE0_W_PMS Configures write permission for TRACE0 to access 32 address ranges.
Bit 0 corresponds to region0, and so on.
0: Disable write permission.
1: Enable write permission.
(R/W)

Register 19.45. PMS_DMA_TRACE1_PMS_R_REG (0x01D4)

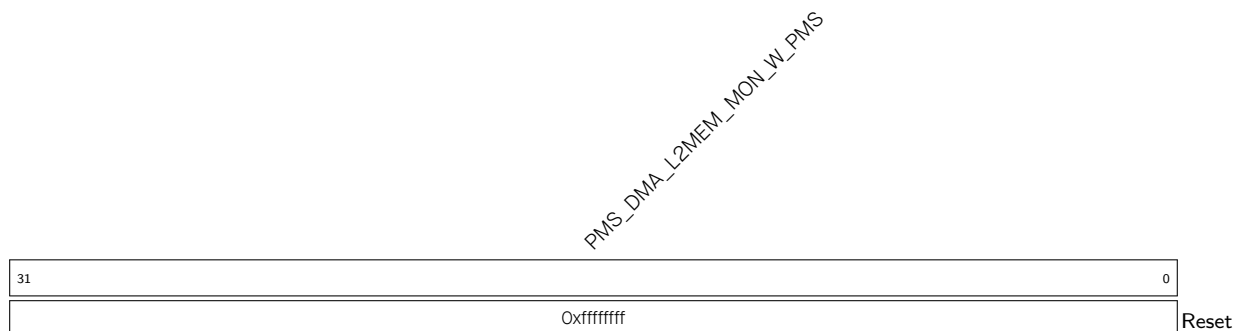


PMS_DMA_TRACE1_R_PMS Configures read permission for TRACE1 to access 32 address ranges.
Bit 0 corresponds to region0, and so on.
0: Disable read permission.
1: Enable read permission.
(R/W)

Register 19.46. PMS_DMA_TRACE1_PMS_W_REG (0x01D8)



PMS_DMA_TRACE1_W_PMS Configures write permission for TRACE1 to access 32 address ranges.
Bit 0 corresponds to region0, and so on.
0: Disable write permission.
1: Enable write permission.
(R/W)

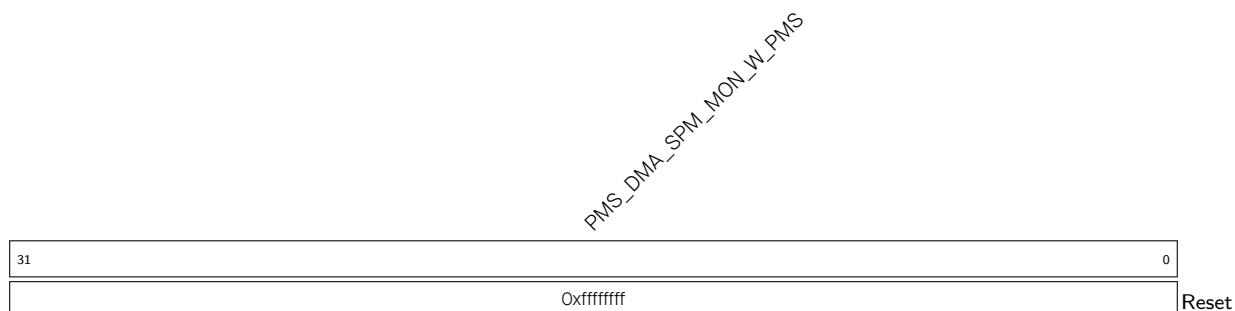
Register 19.47. PMS_DMA_L2MEM_MON_PMS_W_REG (0x01E0)

PMS_DMA_L2MEM_MON_W_PMS Configures write permission for L2MEM monitor to access 32 address ranges. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

(R/W)

Register 19.48. PMS_DMA_SPM_MON_PMS_W_REG (0x01E8)

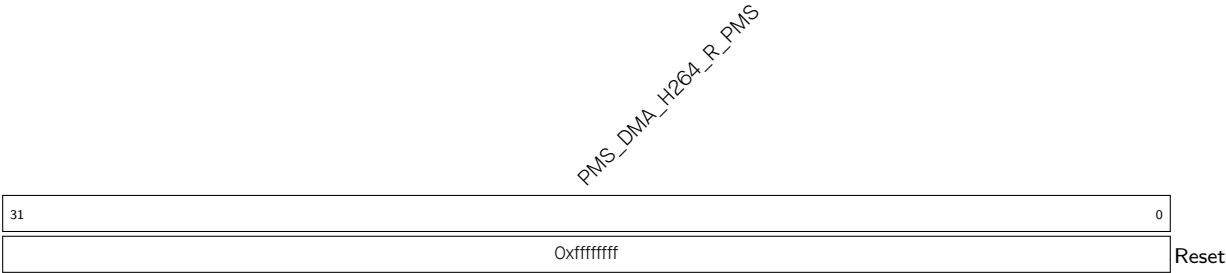
PMS_DMA_SPM_MON_W_PMS Configures write permission for SPM monitor to access 32 address ranges. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

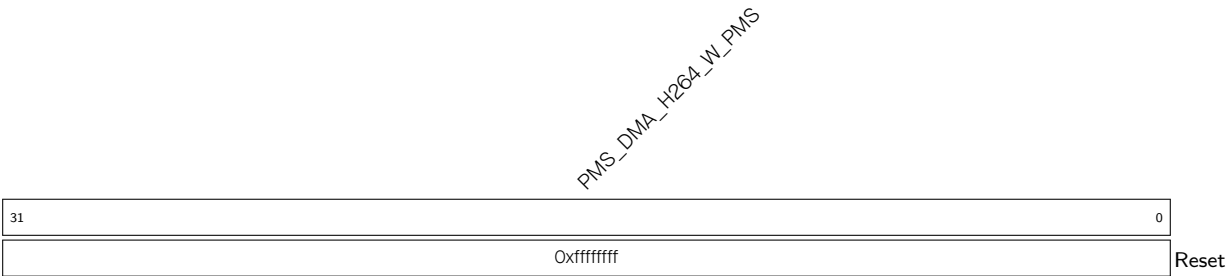
(R/W)

Register 19.49. PMS_DMA_H264_PMS_R_REG (0x01FC)

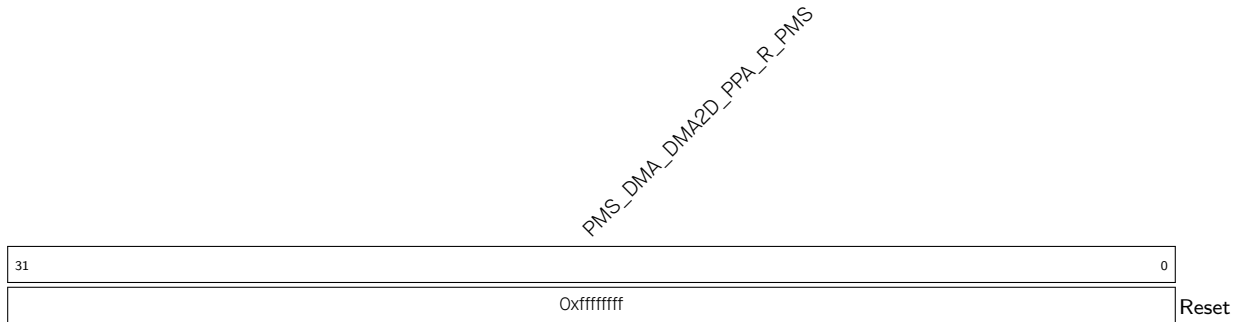


PMS_DMA_H264_R_PMS Configures read permission for H264 DMA to access 32 address ranges.
Bit 0 corresponds to region0, and so on.
0: Disable read permission.
1: Enable read permission.
(R/W)

Register 19.50. PMS_DMA_H264_PMS_W_REG (0x0200)



PMS_DMA_H264_W_PMS Configures write permission for H264 DMA to access 32 address ranges.
Bit 0 corresponds to region0, and so on.
0: Disable write permission.
1: Enable write permission.
(R/W)

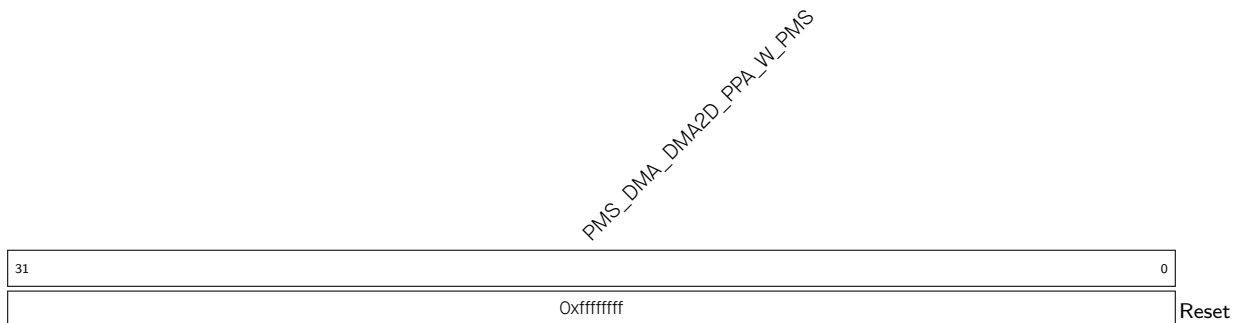
Register 19.51. PMS_DMA_DMA2D_PPA_PMS_R_REG (0x0204)

PMS_DMA_DMA2D_PPA_R_PMS Configures 2D-DMA permission to read 32 address ranges requested by PPA (Pixel-Processing Accelerator). Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

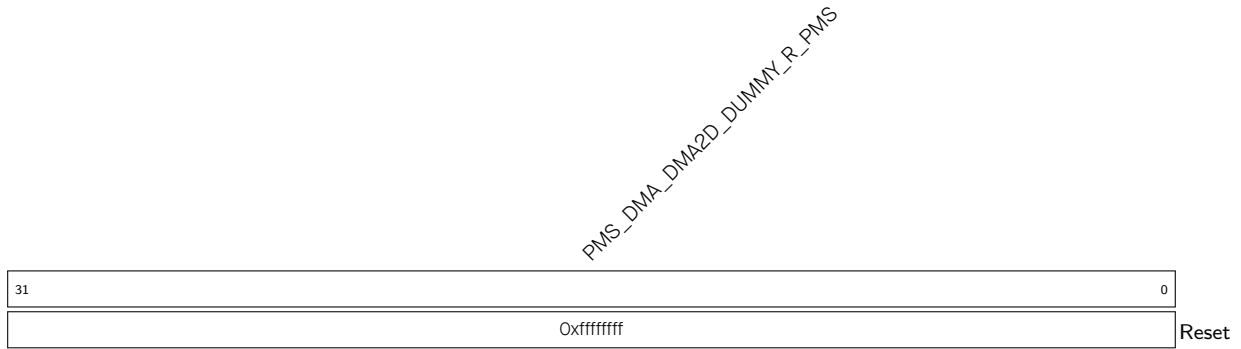
Register 19.52. PMS_DMA_DMA2D_PPA_PMS_W_REG (0x0208)

PMS_DMA_DMA2D_PPA_W_PMS Configures 2D-DMA permission to write 32 address ranges requested by PPA. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

(R/W)

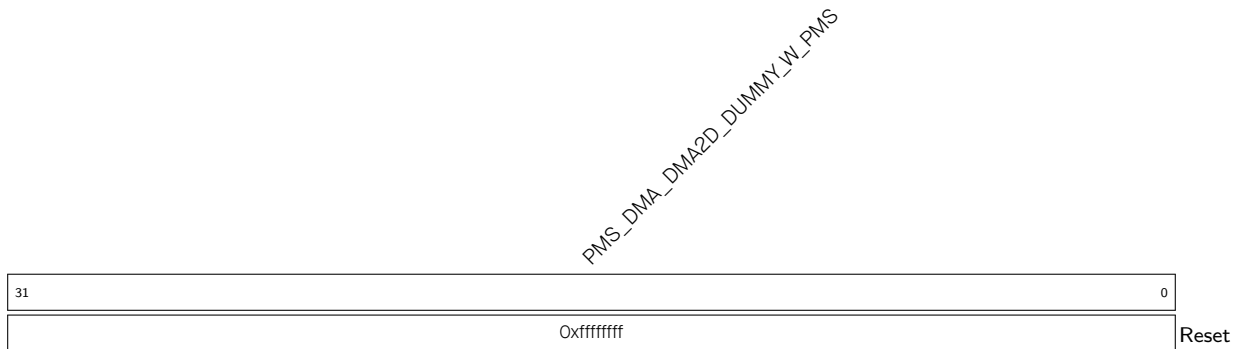
Register 19.53. PMS_DMA_DMA2D_DUMMY_PMS_R_REG (0x020C)

PMS_DMA_DMA2D_DUMMY_R_PMS Configures 2D-DMA permission to read 32 address ranges requested by Dummy. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

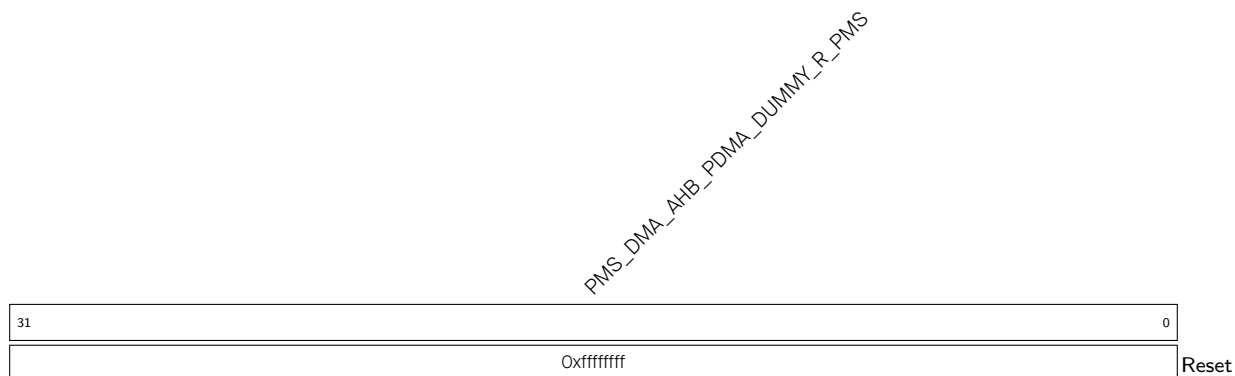
Register 19.54. PMS_DMA_DMA2D_DUMMY_PMS_W_REG (0x0210)

PMS_DMA_DMA2D_DUMMY_W_PMS Configures 2D-DMA permission to write 32 address ranges requested by Dummy. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

(R/W)

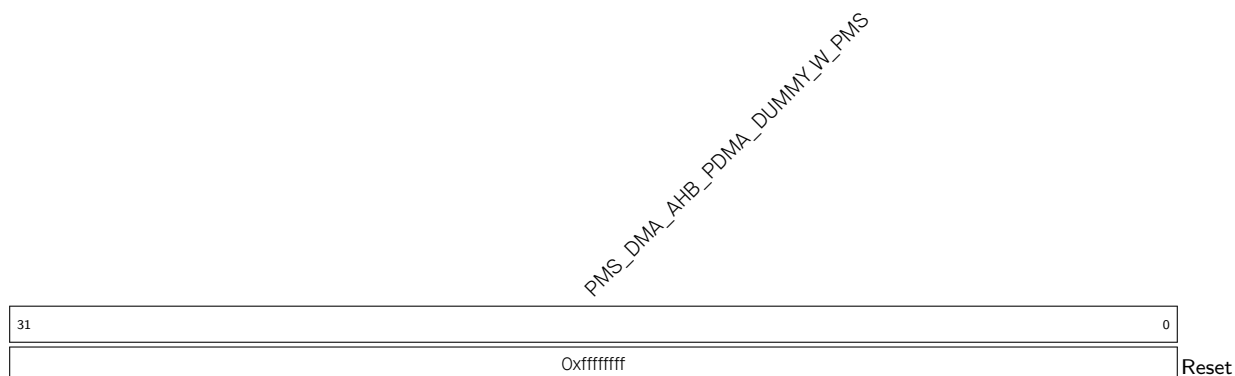
Register 19.55. PMS_DMA_AHB_PDMA_DUMMY_R_PMS_REG (0x0214)

PMS_DMA_AHB_PDMA_DUMMY_R_PMS Configures GDMA-AHB permission to read 32 address ranges requested by Dummy. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

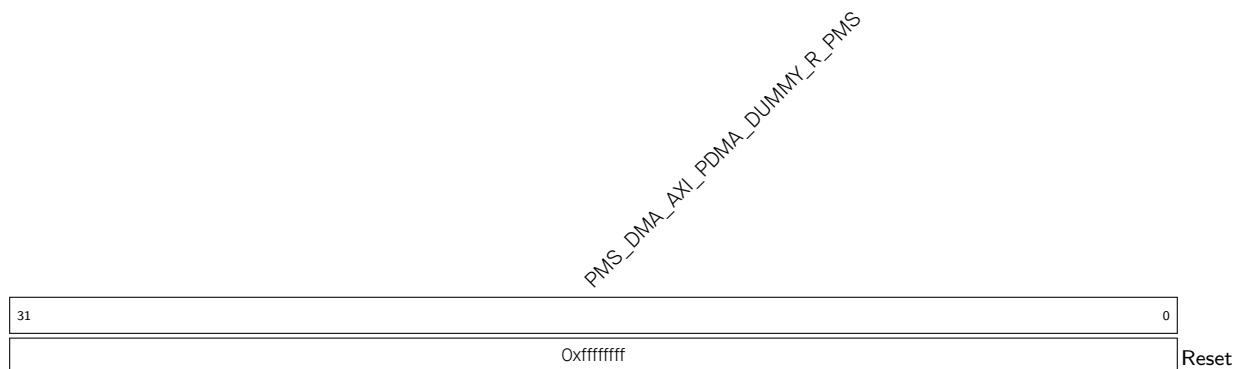
Register 19.56. PMS_DMA_AHB_PDMA_DUMMY_W_PMS_REG (0x0218)

PMS_DMA_AHB_PDMA_DUMMY_W_PMS Configures GDMA-AHB permission to write 32 address ranges requested by Dummy. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

(R/W)

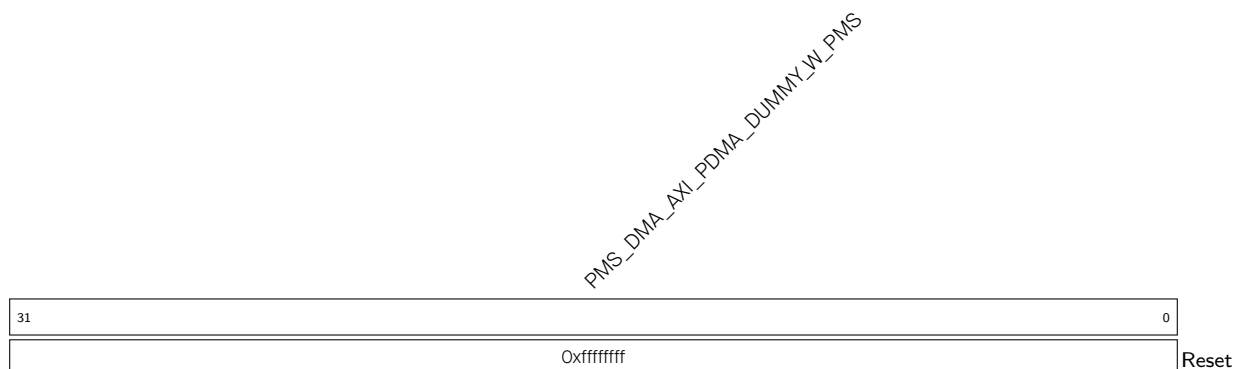
Register 19.57. PMS_DMA_AXI_PDMA_DUMMY_R_PMS_REG (0x021C)

PMS_DMA_AXI_PDMA_DUMMY_R_PMS Configures GDMA-AXI permission to read 32 address ranges requested by Dummy. Bit 0 corresponds to region0, and so on.

0: Disable read permission.

1: Enable read permission.

(R/W)

Register 19.58. PMS_DMA_AXI_PDMA_DUMMY_W_PMS_REG (0x0220)

PMS_DMA_AXI_PDMA_DUMMY_W_PMS Configures GDMA-AXI permission to write 32 address ranges requested by Dummy. Bit 0 corresponds to region0, and so on.

0: Disable write permission.

1: Enable write permission.

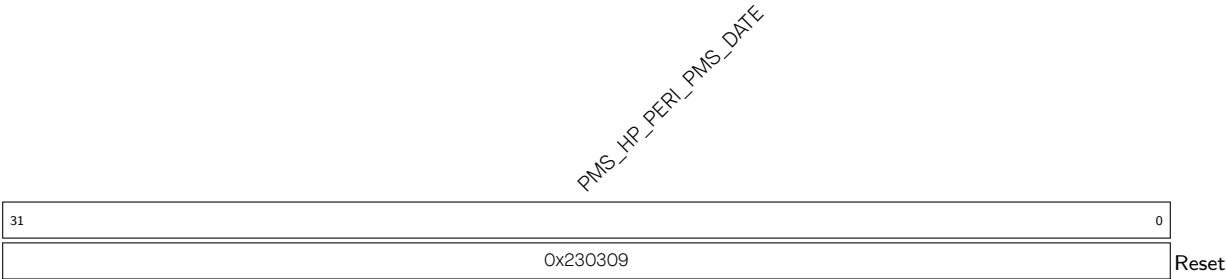
(R/W)

19.7.2 HP_PERI_PMS_REG

The addresses in this section are relative to the HP_PERI_PMS base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

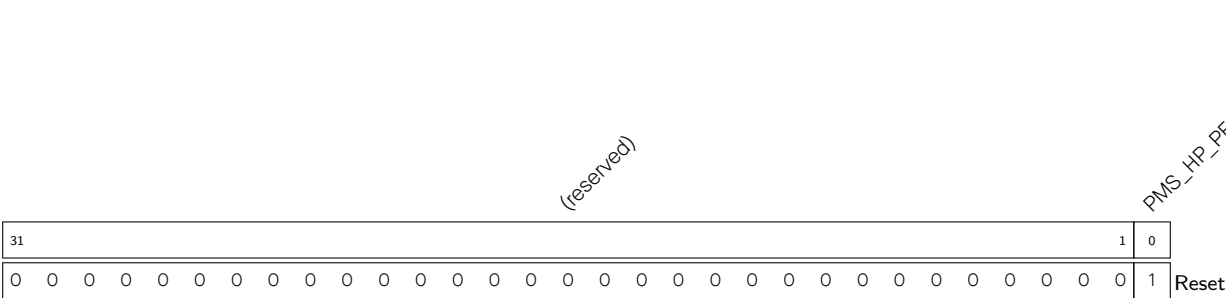
For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 19.59. PMS_HP_PERI_PMS_DATE_REG (0x0000)



PMS_HP_PERI_PMS_DATE Version control register. (R/W)

Register 19.60. PMS_HP_PERI_PMS_CLK_EN_REG (0x0004)



PMS_HP_PERI_PMS_CLK_EN Configures whether to keep the clock always on.

- 0: Enable automatic clock gating
- 1: Keep the clock always on

(R/W)

Register 19.61. PMS_CORE n _MM_HP_PERI_PMS_REGO_REG (n : 0-1) (0x0008+0x20* n)

(reserved)																				PMS_CORE n _MM_CACHE_ALLOW PMS_CORE n _MM_SPM_MON_ALLOW PMS_CORE n _MM_L2MEM_MON_ALLOW PMS_CORE n _MM_CPU_BUS_MON_ALLOW PMS_CORE n _MM_TRACE1_ALLOW PMS_CORE n _MM_TRACE0_ALLOW (reserved) PMS_CORE n _MM_L2ROM_ALLOW PMS_CORE n _MM_L2MEM_ALLOW PMS_CORE n _MM_FLASH_ALLOW PMS_CORE n _MM_PSRAM_ALLOW													
31																				12	11	10	9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	1	1	1	1					

Reset

PMS_CORE n _MM_PSRAM_ALLOW Configures whether HP CPU n in machine mode has permission to access external RAM without going through cache.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_FLASH_ALLOW Configures whether HP CPU n in machine mode has permission to access external flash without going through cache.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_L2MEM_ALLOW Configures whether HP CPU n in machine mode has permission to access HP L2MEM without going through cache.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_L2ROM_ALLOW Configures whether HP CPU n in machine mode has permission to access HP ROM without going through cache.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_TRACE0_ALLOW Configures whether HP CPU n in machine mode has permission to access TRACE0.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_TRACE1_ALLOW Configures whether HP CPU n in machine mode has permission to access TRACE1.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.61. PMS_CORE n _MM_HP_PERI_PMS_REGO_REG (n : 0-1) (0x0008+0x20* n)

Continued from the previous page...

PMS_CORE n _MM_CPU_BUS_MON_ALLOW Configures whether HP CPU n in machine mode has permission to access CPU bus monitor.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_L2MEM_MON_ALLOW Configures whether HP CPU n in machine mode has permission to access L2MEM monitor.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_SPM_MON_ALLOW Configures whether HP CPU n in machine mode has permission to access SPM monitor.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_CACHE_ALLOW Configures whether HP CPU n in machine mode has permission to access cache.

0: Not allowed

1: Allowed

(R/W)

Register 19.62. PMS_CORE n _MM_HP_PERI_PMS_REG1_REG (n : 0-1) (0x000C+0x20* n)

(reserved) PMS_COREn_MM_HP_H264_DMA2D_ALLOW PMS_COREn_MM_DMA_PMS_ALLOW PMS_COREn_MM_LP2HP_PERI_PMS_ALLOW PMS_COREn_MM_HP_PERI_PMS_ALLOW PMS_COREn_MM_HP_AXI_ICM_ALLOW PMS_COREn_MM_HP_BITSCRAMBLER_ALLOW PMS_COREn_MM_HP_RMT_ALLOW PMS_COREn_MM_HP_H264_CORE_ALLOW (reserved) PMS_COREn_MM_HP_ISP_ALLOW PMS_COREn_MM_HP_DSI_HOST_ALLOW PMS_COREn_MM_HP_CSI_HOST_ALLOW PMS_COREn_MM_HP_USB_PHY_ALLOW PMS_COREn_MM_HP_GMAC_ALLOW PMS_COREn_MM_HP_CRYPTO_ALLOW PMS_COREn_MM_HP_PSRAM_ALLOW PMS_COREn_MM_HP_FLASH_ALLOW PMS_COREn_MM_HP_AXI_PDMA_ALLOW PMS_COREn_MM_HP_KEY_MANAGER_ALLOW PMS_COREn_MM_HP_DMA2D_ALLOW PMS_COREn_MM_HP_PPA_ALLOW PMS_COREn_MM_HP_JPEG_ALLOW (reserved) PMS_COREn_MM_HP_AHB_PDMA_ALLOW PMS_COREn_MM_HP_SDMMC_ALLOW PMS_COREn_MM_HP_GDMA_ALLOW PMS_COREn_MM_HP_USBOTG11_WRAP_ALLOW PMS_COREn_MM_HP_USBOTG11_ALLOW PMS_COREn_MM_HP_USBOTG_ALLOW																															
31	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
Reset																															

Reset

PMS_CORE n _MM_HP_USBOTG_ALLOW Configures whether HP CPU n in machine mode has permission to access HP high-speed USB 2.0 OTG.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_USBOTG11_ALLOW Configures whether HP CPU n in machine mode has permission to access HP full-speed USB 2.0 OTG.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_USBOTG11_WRAP_ALLOW Configures whether HP CPU n in machine mode has permission to access HP full-speed USB 2.0 OTG's wrap.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_GDMA_ALLOW Configures whether HP CPU n in machine mode has permission to access HP VDMA.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_SDMMC_ALLOW Configures whether HP CPU n in machine mode has permission to access HP SDMMC.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.62. PMS_CORE n _MM_HP_PERI_PMS_REG1_REG (n : 0-1) (0x000C+0x20* n)

Continued from the previous page...

PMS_CORE n _MM_HP_AHB_PDMA_ALLOW Configures whether HP CPU n in machine mode has permission to access GDMA-AHB.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_JPEG_ALLOW Configures whether HP CPU n in machine mode has permission to access HP JPEG.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_PPA_ALLOW Configures whether HP CPU n in machine mode has permission to access HP PPA.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_DMA2D_ALLOW Configures whether HP CPU n in machine mode has permission to access HP 2D-DMA.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_KEY_MANAGER_ALLOW Configures whether HP CPU n in machine mode has permission to access HP Key Manager.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_AXI_PDMA_ALLOW Configures whether HP CPU n in machine mode has permission to access HP GDMA-AXI.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_FLASH_ALLOW Configures whether HP CPU n in machine mode has permission to access HP flash MSPI controller.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.62. PMS_CORE n _MM_HP_PERI_PMS_REG1_REG (n : 0-1) (0x000C+0x20* n)

Continued from the previous page...

PMS_CORE n _MM_HP_PSRAM_ALLOW Configures whether HP CPU n in machine mode has permission to access HP PSRAM MSPI controller.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_CRYPTO_ALLOW Configures whether HP CPU n in machine mode has permission to access HP CRYPTO (including AES/SHA/RSA/HMAC Accelerators).

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_GMAC_ALLOW Configures whether HP CPU n in machine mode has permission to access HP EMAC.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_USB_PHY_ALLOW Configures whether HP CPU n in machine mode has permission to access HP high-speed USB 2.0 OTG PHY.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_CSI_HOST_ALLOW Configures whether HP CPU n in machine mode has permission to access HP MIPI CSI host.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_DSI_HOST_ALLOW Configures whether HP CPU n in machine mode has permission to access HP MIPI DSI host.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_ISP_ALLOW Configures whether HP CPU n in machine mode has permission to access HP ISP (Image Signal Processor).

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.62. PMS_CORE n _MM_HP_PERI_PMS_REG1_REG (n : 0-1) (0x000C+0x20* n)

Continued from the previous page...

PMS_CORE n _MM_HP_H264_CORE_ALLOW Configures whether HP CPU n in machine mode has permission to access HP H264 Encoder.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_RMT_ALLOW Configures whether HP CPU n in machine mode has permission to access HP RMT.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_BITSCRAMBLER_ALLOW Configures whether HP CPU n in machine mode has permission to access HP bit scrambler.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_AXI_ICM_ALLOW Configures whether HP CPU n in machine mode has permission to access HP AXI ICM.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_PERI_PMS_ALLOW Configures whether HP CPU n in machine mode has permission to access HP_PERI_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_LP2HP_PERI_PMS_ALLOW Configures whether HP CPU n in machine mode has permission to access LP2HP_PERI_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_DMA_PMS_ALLOW Configures whether HP CPU n in machine mode has permission to access HP_DMA_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_H264_DMA2D_ALLOW Configures whether HP CPU n in machine mode has permission to access 2D-DMA.

0: Not allowed

1: Allowed

(R/W)

Register 19.63. PMS_CORE n _MM_HP_PERI_PMS_REG2_REG (n : 0-1) (0x0010+0x20* n)

PMS_COREn_MM_HP_UHCI_ALLOW																															
PMS_COREn_MM_HP_ADC_ALLOW																															
(reserved)																															
PMS_COREn_MM_HP_LCDAM_ALLOW																															
PMS_COREn_MM_HP_I3C_SIV_ALLOW																															
PMS_COREn_MM_HP_I3C_MST_ALLOW																															
PMS_COREn_MM_HP_TWI2_ALLOW																															
PMS_COREn_MM_HP_TWI1_ALLOW																															
(reserved)																															
PMS_COREn_MM_HP_INTRMTX_ALLOW																															
PMS_COREn_MM_HP_ETM_ALLOW																															
PMS_COREn_MM_HP_LEDC_ALLOW																															
PMS_COREn_MM_HP_USBDEVICE_ALLOW																															
PMS_COREn_MM_HP_GSP13_ALLOW																															
PMS_COREn_MM_HP_GSP12_ALLOW																															
PMS_COREn_MM_HP_PARLIO_ALLOW																															
PMS_COREn_MM_HP_UART4_ALLOW																															
PMS_COREn_MM_HP_UART3_ALLOW																															
PMS_COREn_MM_HP_UART2_ALLOW																															
PMS_COREn_MM_HP_UART1_ALLOW																															
PMS_COREn_MM_HP_PCNT_ALLOW																															
PMS_COREn_MM_HP_I2S2_ALLOW																															
PMS_COREn_MM_HP_I2S1_ALLOW																															
PMS_COREn_MM_HP_I2S0_ALLOW																															
PMS_COREn_MM_HP_I2C1_ALLOW																															
PMS_COREn_MM_HP_I2C0_ALLOW																															
PMS_COREn_MM_HP_TIMER_GROUP1_ALLOW																															
PMS_COREn_MM_HP_TIMER_GROUP0_ALLOW																															
PMS_COREn_MM_HP_MCPWM1_ALLOW																															
PMS_COREn_MM_HP_MCPWM0_ALLOW																															

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
1	1	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	Reset

PMS_CORE n _MM_HP_MCPWM0_ALLOW Configures whether HP CPU n in machine mode has permission to access HP MCPWM0.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_MCPWM1_ALLOW Configures whether HP CPU n in machine mode has permission to access HP MCPWM1.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_TIMER_GROUP0_ALLOW Configures whether HP CPU n in machine mode has permission to access HP timer group0.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_TIMER_GROUP1_ALLOW Configures whether HP CPU n in machine mode has permission to access HP timer group1.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_I2C0_ALLOW Configures whether HP CPU n in machine mode has permission to access HP I2C0.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_I2C1_ALLOW Configures whether HP CPU n in machine mode has permission to access HP I2C1.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.63. PMS_CORE n _MM_HP_PERI_PMS_REG2_REG (n : 0-1) (0x000C+0x20* n)

Continued from the previous page...

PMS_CORE n _MM_HP_I2S0_ALLOW Configures whether HP CPU n in machine mode has permission to access HP I2S0.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_I2S1_ALLOW Configures whether HP CPU n in machine mode has permission to access HP I2S1.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_I2S2_ALLOW Configures whether HP CPU n in machine mode has permission to access HP I2S2.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_PCNT_ALLOW Configures whether HP CPU n in machine mode has permission to access HP PCNT.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_UART0_ALLOW Configures whether HP CPU n in machine mode has permission to access HP UART0.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_UART1_ALLOW Configures whether HP CPU n in machine mode has permission to access HP UART1.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_UART2_ALLOW Configures whether HP CPU n in machine mode has permission to access HP UART2.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.63. PMS_CORE n _MM_HP_PERI_PMS_REG2_REG (n : 0-1) (0x000C+0x20* n)

Continued from the previous page...

PMS_CORE n _MM_HP_UART3_ALLOW Configures whether HP CPU n in machine mode has permission to access HP UART3.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_UART4_ALLOW Configures whether HP CPU n in machine mode has permission to access HP UART4.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_PARLIO_ALLOW Configures whether HP CPU n in machine mode has permission to access HP PARLIO.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_GPSPI2_ALLOW Configures whether HP CPU n in machine mode has permission to access HP GP-SPI2.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_GPSPI3_ALLOW Configures whether HP CPU n in machine mode has permission to access HP GP-SPI3.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_USBDEVICE_ALLOW Configures whether HP CPU n in machine mode has permission to access HP USB Serial/JTAG Controller.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_LEDC_ALLOW Configures whether HP CPU n in machine mode has permission to access HP LEDC.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.63. PMS_CORE n _MM_HP_PERI_PMS_REG2_REG (n : 0-1) (0x000C+0x20* n)

Continued from the previous page...

PMS_CORE n _MM_HP_ETM_ALLOW Configures whether HP CPU n in machine mode has permission to access HP ETM (Event Task Matrix).

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_INTRMTX_ALLOW Configures whether HP CPU n in machine mode has permission to access HP interrupt matrix.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_TWAIO_ALLOW Configures whether HP CPU n in machine mode has permission to access HP TWAIO.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_TWAI1_ALLOW Configures whether HP CPU n in machine mode has permission to access HP TWAI1.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_TWAI2_ALLOW Configures whether HP CPU n in machine mode has permission to access HP TWAI2.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_I3C_MST_ALLOW Configures whether HP CPU n in machine mode has permission to access HP I3C master controller.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_I3C_SLV_ALLOW Configures whether HP CPU n in machine mode has permission to access HP I3C slave controller.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.63. PMS_CORE n _MM_HP_PERI_PMS_REG2_REG (n : 0-1) (0x000C+0x20* n)

Continued from the previous page...

PMS_CORE n _MM_HP_LCDCAM_ALLOW Configures whether HP CPU n in machine mode has permission to access HP LCD_CAM.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_ADC_ALLOW Configures whether HP CPU n in machine mode has permission to access HP ADC.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_UHCI_ALLOW Configures whether HP CPU n in machine mode has permission to access HP UHCI.

0: Not allowed

1: Allowed

(R/W)

Register 19.64. PMS_CORE n _MM_HP_PERI_PMS_REG3_REG (n : 0-1) (0x0014+0x20* n)

(reserved)																										PMS_CORE _n _MM_HP_CLKRST_ALLOW					
																										PMS_CORE _n _MM_HP_SYS_REG_ALLOW					
																										PMS_CORE _n _MM_HP_SYSTIMER_ALLOW					
																										PMS_CORE _n _MM_HP_IOMUX_ALLOW					
																										PMS_CORE _n _MM_HP_GPIO_ALLOW					
31																							5	4	3	2	1	0			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	Reset		

PMS_CORE n _MM_HP_GPIO_ALLOW Configures whether HP CPU n in machine mode has permission to access HP GPIO Matrix.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_IOMUX_ALLOW Configures whether HP CPU n in machine mode has permission to access HP IO MUX.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_SYSTIMER_ALLOW Configures whether HP CPU n in machine mode has permission to access HP system timer.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_SYS_REG_ALLOW Configures whether HP CPU n in machine mode has permission to access HP system register.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _MM_HP_CLKRST_ALLOW Configures whether HP CPU n in machine mode has permission to access HP_SYS_CLKRST.

0: Not allowed

1: Allowed

(R/W)

Register 19.65. PMS_CORE n _UM_HP_PERI_PMS_REGO_REG (n : 0-1) (0x0018+0x20* n)

(reserved)																								PMS_CORE_n_UM_CACHE_ALLOW PMS_CORE_n_UM_SPM_MON_ALLOW PMS_CORE_n_UM_L2MEM_MON_ALLOW PMS_CORE_n_UM_CPU_BUS_MON_ALLOW PMS_CORE_n_UM_TRACE1_ALLOW PMS_CORE_n_UM_TRACE0_ALLOW (reserved) PMS_CORE_n_UM_L2FROM_ALLOW PMS_CORE_n_UM_L2MEM_ALLOW PMS_CORE_n_UM_FLASH_ALLOW PMS_CORE_n_UM_PSRAM_ALLOW																													
31																								12								11	10	9	8	7	6	5	4	3	2	1	0										
0																								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	0	1	1	1	1	Reset

PMS_CORE n _UM_PSRAM_ALLOW Configures whether HP CPU n in user mode has permission to access external RAM without going through cache.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_FLASH_ALLOW Configures whether HP CPU n in user mode has permission to access external flash without going through cache.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_L2MEM_ALLOW Configures whether HP CPU n in user mode has permission to access HP L2MEM without going through cache.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_L2ROM_ALLOW Configures whether HP CPU n in user mode has permission to access HP ROM without going through cache.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_TRACE0_ALLOW Configures whether HP CPU n in user mode has permission to access TRACE0.
0: Not allowed
1: Allowed
(R/W)

PMS_CORE n _UM_TRACE1_ALLOW Configures whether HP CPU n in user mode has permission to access TRACE1.
 0: Not allowed
 1: Allowed
 (R/W)

Continued on the next page...

Register 19.65. PMS_CORE n _UM_HP_PERI_PMS_REGO_REG (n : 0-1) (0x000C+0x20* n)

Continued from the previous page...

PMS_CORE n _UM_CPU_BUS_MON_ALLOW Configures whether HP CPU n in user mode has permission to access CPU bus monitor.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_L2MEM_MON_ALLOW Configures whether HP CPU n in user mode has permission to access L2MEM monitor.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_SPM_MON_ALLOW Configures whether HP CPU n in user mode has permission to access SPM monitor.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_CACHE_ALLOW Configures whether HP CPU n in user mode has permission to access cache.

0: Not allowed

1: Allowed

(R/W)

Register 19.66. PMS_CORE n _UM_HP_PERI_PMS_REG1_REG (n : 0-1) (0x001C+0x20* n)

(reserved) PMS_COREn_UM_HP_H264_DMA2D_ALLOW PMS_COREn_UM_DMA_PMS_ALLOW PMS_COREn_UM_LP2HP_PERI_PMS_ALLOW PMS_COREn_UM_HP_PERI_PMS_ALLOW PMS_COREn_UM_HP_AXI_ICM_ALLOW PMS_COREn_UM_HP_RMT_ALLOW PMS_COREn_UM_HP_H264_CORE_ALLOW PMS_COREn_UM_HP_ISP_ALLOW PMS_COREn_UM_HP_DSI_HOST_ALLOW PMS_COREn_UM_HP_CSI_HOST_ALLOW PMS_COREn_UM_HP_USB_PHY_ALLOW PMS_COREn_UM_HP_GMAC_ALLOW PMS_COREn_UM_HP_CRYPTO_ALLOW PMS_COREn_UM_HP_PSRAM_ALLOW PMS_COREn_UM_HP_FLASH_ALLOW PMS_COREn_UM_HP_AXI_PDMA_ALLOW PMS_COREn_UM_HP_KEY_MANAGER_ALLOW PMS_COREn_UM_HP_DMA2D_ALLOW PMS_COREn_UM_HP_PPA_ALLOW PMS_COREn_UM_HP_JPEG_ALLOW (reserved) PMS_COREn_UM_HP_AHB_PDMA_ALLOW PMS_COREn_UM_HP_SDMMC_ALLOW PMS_COREn_UM_HP_GDMA_ALLOW PMS_COREn_UM_HP_USBOTG1_WRAP_ALLOW PMS_COREn_UM_HP_USBOTG1_ALLOW PMS_COREn_UM_HP_USBOTG_ALLOW																															
31	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	Reset

Reset

PMS_CORE n _UM_HP_USBOTG_ALLOW Configures whether HP CPU n in user mode has permission to access HP high-speed USB 2.0 OTG.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_USBOTG11_ALLOW Configures whether HP CPU n in user mode has permission to access HP full-speed USB 2.0 OTG.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_USBOTG11_WRAP_ALLOW Configures whether HP CPU n in user mode has permission to access HP full-speed USB 2.0 OTG's wrap.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_GDMA_ALLOW Configures whether HP CPU n in user mode has permission to access HP VDMA.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_SDMMC_ALLOW Configures whether HP CPU n in user mode has permission to access HP SDMMC.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.66. PMS_CORE n _UM_HP_PERI_PMS_REG1_REG (n : 0-1) (0x000C+0x20* n)

Continued from the previous page...

PMS_CORE n _UM_HP_AHB_PDMA_ALLOW Configures whether HP CPU n in user mode has permission to access GDMA-AHB.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_JPEG_ALLOW Configures whether HP CPU n in user mode has permission to access HP JPEG.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_PPA_ALLOW Configures whether HP CPU n in user mode has permission to access HP PPA.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_DMA2D_ALLOW Configures whether HP CPU n in user mode has permission to access HP 2D-DMA.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_KEY_MANAGER_ALLOW Configures whether HP CPU n in user mode has permission to access HP Key Manager.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_AXI_PDMA_ALLOW Configures whether HP CPU n in user mode has permission to access HP GDMA-AXI.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_FLASH_ALLOW Configures whether HP CPU n in user mode has permission to access HP flash MSPI controller.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_PSRAM_ALLOW Configures whether HP CPU n in user mode has permission to access HP PSRAM MSPI controller.

0: Not allowed

1: Allowed

(R/W)

Register 19.66. PMS_CORE n _UM_HP_PERI_PMS_REG1_REG (n : 0-1) (0x000C+0x20* n)

Continued from the previous page...

PMS_CORE n _UM_HP_CRYPTO_ALLOW Configures whether HP CPU n in user mode has permission to access HP CRYPTO.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_GMAC_ALLOW Configures whether HP CPU n in user mode has permission to access HP EMAC.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_USB_PHY_ALLOW Configures whether HP CPU n in user mode has permission to access HP high-speed USB 2.0 OTG PHY.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_CSI_HOST_ALLOW Configures whether HP CPU n in user mode has permission to access HP MIPI CSI host.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_DSI_HOST_ALLOW Configures whether HP CPU n in user mode has permission to access HP MIPI DSI host.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_ISP_ALLOW Configures whether HP CPU n in user mode has permission to access HP ISP.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_H264_CORE_ALLOW Configures whether HP CPU n in user mode has permission to access HP H264 Encoder.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.66. PMS_CORE n _UM_HP_PERI_PMS_REG1_REG (n : 0-1) (0x000C+0x20* n)

Continued from the previous page...

PMS_CORE n _UM_HP_RMT_ALLOW Configures whether HP CPU n in user mode has permission to access HP RMT.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_BITSCRAMBLER_ALLOW Configures whether HP CPU n in user mode has permission to access HP bit scrambler.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_AXI_ICM_ALLOW Configures whether HP CPU n in user mode has permission to access HP AXI ICM.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_PERI_PMS_ALLOW Configures whether HP CPU n in user mode has permission to access HP_PERI_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_LP2HP_PERI_PMS_ALLOW Configures whether HP CPU n in user mode has permission to access LP2HP_PERI_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_DMA_PMS_ALLOW Configures whether HP CPU n in user mode has permission to access HP_DMA_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_H264_DMA2D_ALLOW Configures whether HP CPU n in user mode has permission to access 2D-DMA.

0: Not allowed

1: Allowed

(R/W)

Register 19.67. PMS_CORE n _UM_HP_PERI_PMS_REG2_REG (n : 0-1) (0x0020+0x20* n)

PMS_CORE _n _UM_HP_UHCL_ALLOW																																PMS_CORE _n _UM_HP_ADC_ALLOW																																(reserved)																																PMS_CORE _n _UM_HP_LCDAM_ALLOW																																PMS_CORE _n _UM_HP_I3C_SIV_ALLOW																																PMS_CORE _n _UM_HP_I3C_MST_ALLOW																																PMS_CORE _n _UM_HP_TWA2_ALLOW																																PMS_CORE _n _UM_HP_TWA1_ALLOW																																(reserved)																																PMS_CORE _n _UM_HP_INTRMTX_ALLOW																																PMS_CORE _n _UM_HP_ETM_ALLOW																																PMS_CORE _n _UM_HP_LEDC_ALLOW																																PMS_CORE _n _UM_HP_USBDDEVICE_ALLOW																																PMS_CORE _n _UM_HP_GPSP13_ALLOW																																PMS_CORE _n _UM_HP_GPSP12_ALLOW																																PMS_CORE _n _UM_HP_PARLIO_ALLOW																																PMS_CORE _n _UM_HP_UART4_ALLOW																																PMS_CORE _n _UM_HP_UART3_ALLOW																																PMS_CORE _n _UM_HP_UART2_ALLOW																																PMS_CORE _n _UM_HP_UART1_ALLOW																																PMS_CORE _n _UM_HP_PCNT_ALLOW																																PMS_CORE _n _UM_HP_I2S2_ALLOW																																PMS_CORE _n _UM_HP_I2S1_ALLOW																																PMS_CORE _n _UM_HP_I2S0_ALLOW																																PMS_CORE _n _UM_HP_I2C1_ALLOW																																PMS_CORE _n _UM_HP_I2C0_ALLOW																																PMS_CORE _n _UM_HP_TIMER_GROUP1_ALLOW																																PMS_CORE _n _UM_HP_TIMER_GROUP0_ALLOW																																PMS_CORE _n _UM_HP_MCPWM1_ALLOW																																PMS_CORE _n _UM_HP_MCPWM0_ALLOW																															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
1	1	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	Reset																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													

PMS_CORE n _UM_HP_MCPWM0_ALLOW Configures whether HP CPU n in user mode has permission to access HP MCPWM0.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_MCPWM1_ALLOW Configures whether HP CPU n in user mode has permission to access HP MCPWM1.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_TIMER_GROUP0_ALLOW Configures whether HP CPU n in user mode has permission to access HP timer group0.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_TIMER_GROUP1_ALLOW Configures whether HP CPU n in user mode has permission to access HP timer group1.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_I2C0_ALLOW Configures whether HP CPU n in user mode has permission to access HP I2C0.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_I2C1_ALLOW Configures whether HP CPU n in user mode has permission to access HP I2C1.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.67. PMS_CORE n _UM_HP_PERI_PMS_REG2_REG (n : 0-1) (0x0020+0x20* n)

Continued from the previous page...

PMS_CORE n _UM_HP_I2S0_ALLOW Configures whether HP CPU n in user mode has permission to access HP I2S0.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_I2S1_ALLOW Configures whether HP CPU n in user mode has permission to access HP I2S1.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_I2S2_ALLOW Configures whether HP CPU n in user mode has permission to access HP I2S2.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_PCNT_ALLOW Configures whether HP CPU n in user mode has permission to access HP PCNT.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_UART0_ALLOW Configures whether HP CPU n in user mode has permission to access HP UART0.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_UART1_ALLOW Configures whether HP CPU n in user mode has permission to access HP UART1.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_UART2_ALLOW Configures whether HP CPU n in user mode has permission to access HP UART2.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.67. PMS_CORE n _UM_HP_PERI_PMS_REG2_REG (n : 0-1) (0x0020+0x20* n)

Continued from the previous page...

PMS_CORE n _UM_HP_UART3_ALLOW Configures whether HP CPU n in user mode has permission to access HP UART3.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_UART4_ALLOW Configures whether HP CPU n in user mode has permission to access HP UART4.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_PARLIO_ALLOW Configures whether HP CPU n in user mode has permission to access HP PARLIO.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_GPSPI2_ALLOW Configures whether HP CPU n in user mode has permission to access HP GP-SPI2.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_GPSPI3_ALLOW Configures whether HP CPU n in user mode has permission to access HP GP-SPI3.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_USBDEVICE_ALLOW Configures whether HP CPU n in user mode has permission to access HP USB/Serial JTAG Controller.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_LEDC_ALLOW Configures whether HP CPU n in user mode has permission to access HP LEDC.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.67. PMS_CORE n _UM_HP_PERI_PMS_REG2_REG (n : 0-1) (0x0020+0x20* n)

Continued from the previous page...

PMS_CORE n _UM_HP_ETM_ALLOW Configures whether HP CPU n in user mode has permission to access HP ETM.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_INTRMTX_ALLOW Configures whether HP CPU n in user mode has permission to access HP interrupt matrix.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_TWAI0_ALLOW Configures whether HP CPU n in user mode has permission to access HP TWAI0.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_TWAI1_ALLOW Configures whether HP CPU n in user mode has permission to access HP TWAI1.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_TWAI2_ALLOW Configures whether HP CPU n in user mode has permission to access HP TWAI2.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_I3C_MST_ALLOW Configures whether HP CPU n in user mode has permission to access HP I3C master controller.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_I3C_SLV_ALLOW Configures whether HP CPU n in user mode has permission to access HP I3C slave controller.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.67. PMS_CORE n _UM_HP_PERI_PMS_REG2_REG (n : 0-1) (0x0020+0x20* n)

Continued from the previous page...

PMS_CORE n _UM_HP_LCDCAM_ALLOW Configures whether HP CPU n in user mode has permission to access HP LCD_CAM.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_ADC_ALLOW Configures whether HP CPU n in user mode has permission to access HP ADC.

0: Not allowed

1: Allowed

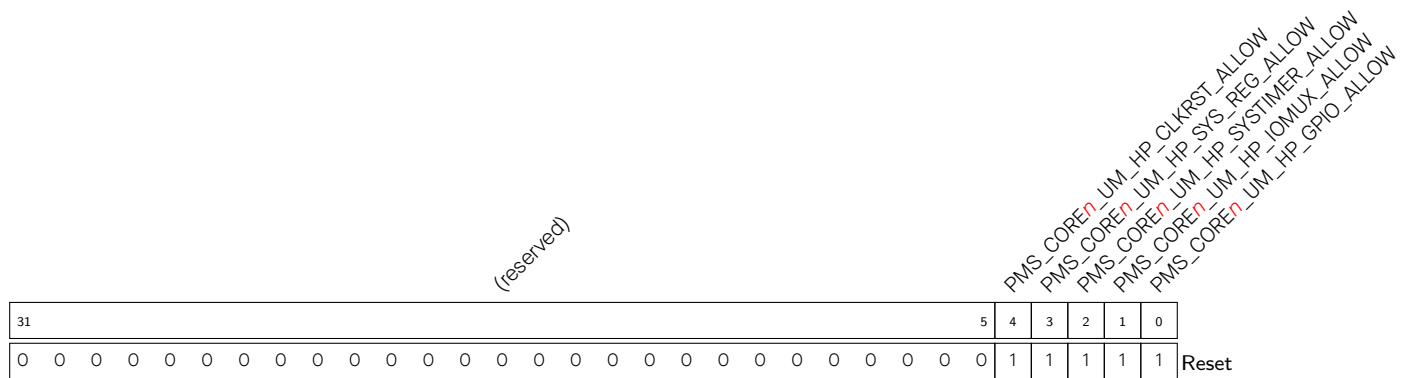
(R/W)

PMS_CORE n _UM_HP_UHCI_ALLOW Configures whether HP CPU n in user mode has permission to access HP UHCI.

0: Not allowed

1: Allowed

(R/W)

Register 19.68. PMS_CORE n _UM_HP_PERI_PMS_REG3_REG (n : 0-1) (0x0024+0x20* n)

PMS_CORE n _UM_HP_GPIO_ALLOW Configures whether HP CPU n in user mode has permission to access HP GPIO Matrix.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_IOMUX_ALLOW Configures whether HP CPU n in user mode has permission to access HP IO MUX.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_SYSTIMER_ALLOW Configures whether HP CPU n in user mode has permission to access HP system timer.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_SYS_REG_ALLOW Configures whether HP CPU n in user mode has permission to access HP system register.

0: Not allowed

1: Allowed

(R/W)

PMS_CORE n _UM_HP_CLKRST_ALLOW Configures whether HP CPU n in user mode has permission to access HP_SYS_CLKRST.

0: Not allowed

1: Allowed

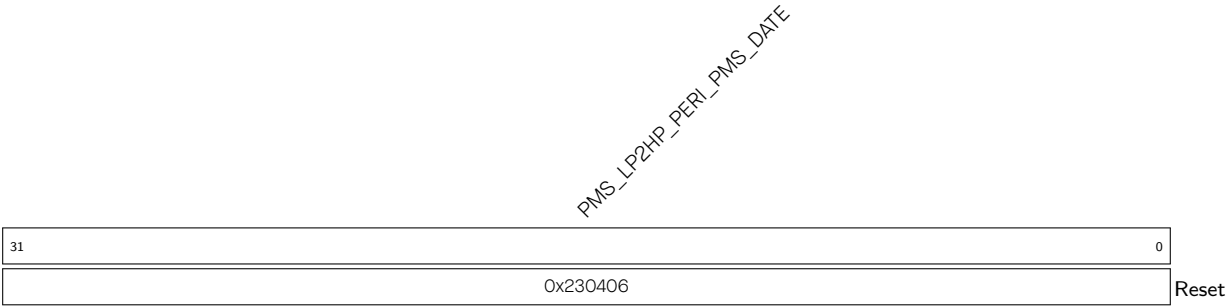
(R/W)

19.7.3 LP2HP_PERI_PMS_REG

The addresses in this section are relative to the LP2HP_PERI_PMS base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

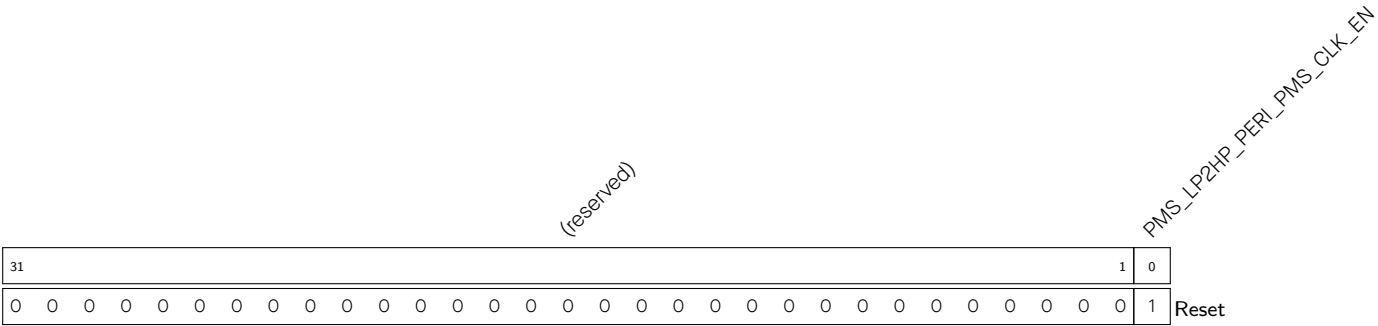
For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 19.69. PMS_LP2HP_PERI_PMS_DATE_REG (0x0000)



PMS_LP2HP_PERI_PMS_DATE Version control register. (R/W)

Register 19.70. PMS_LP2HP_PERI_PMS_CLK_EN_REG (0x0004)



PMS_LP2HP_PERI_PMS_CLK_EN Configures whether to keep the clock always on.

- 0: Enable automatic clock gating.
- 1: Keep the clock always on.

(R/W)

Register 19.71. PMS_LP_MM_PMS_REGO_REG (0x0008)

(reserved)																				PMS_LP_MM_CACHE_ALLOW PMS_LP_MM_SPM_MON_ALLOW PMS_LP_MM_L2MEM_MON_ALLOW PMS_LP_MM_CPU_BUS_MON_ALLOW PMS_LP_MM_TRACE1_ALLOW PMS_LP_MM_TRACE0_ALLOW (reserved) PMS_LP_MM_L2ROM_ALLOW PMS_LP_MM_L2MEM_ALLOW PMS_LP_MM_FLASH_ALLOW PMS_LP_MM_PSRAM_ALLOW											
31												12	11	10	9	8	7	6	5	4	3	2	1	0							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	0	1	1	1	1	Reset

PMS_LP_MM_PSRAM_ALLOW Configures whether the LP CPU in machine mode has permission to access external RAM without going through cache.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_FLASH_ALLOW Configures whether the LP CPU in machine mode has permission to access external flash without going through cache.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_L2MEM_ALLOW Configures whether the LP CPU in machine mode has permission to access HP L2M2M without going through cache.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_L2ROM_ALLOW Configures whether the LP CPU in machine mode has permission to access HP ROM without going through cache.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_TRACE0_ALLOW Configures whether the LP CPU in machine mode has permission to access TRACE0.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_TRACE1_ALLOW Configures whether the LP CPU in machine mode has permission to access TRACE1.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.71. PMS_LP_MM_PMS_REGO_REG (0x0008)

Continued from the previous page...

PMS_LP_MM_CPU_BUS_MON_ALLOW Configures whether the LP CPU in machine mode has permission to access CPU bus monitor.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_L2MEM_MON_ALLOW Configures whether the LP CPU in machine mode has permission to access L2MEM monitor.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_SPM_MON_ALLOW Configures whether the LP CPU in machine mode has permission to access SPM monitor.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_CACHE_ALLOW Configures whether the LP CPU in machine mode has permission to access cache.

0: Not allowed

1: Allowed

(R/W)

Register 19.72. PMS_LP_MM_PMS_REG1_REG (0x0030)

(reserved) PMS_LP_MM_HP_H264_DMA2D_ALLOW PMS_LP_MM_DMA_PMS_ALLOW PMS_LP_MM_LP2HP_PERI_PMS_ALLOW PMS_LP_MM_HP_PERI_PMS_ALLOW PMS_LP_MM_HP_AXI_ICM_ALLOW PMS_LP_MM_HP_BITSCRAMBLER_ALLOW PMS_LP_MM_HP_H264_CORE_ALLOW (reserved) PMS_LP_MM_HP_ISP_ALLOW PMS_LP_MM_HP_DSI_HOST_ALLOW PMS_LP_MM_HP_CSI_HOST_ALLOW PMS_LP_MM_HP_USB_PHY_ALLOW PMS_LP_MM_HP_GMAC_ALLOW PMS_LP_MM_HP_CRYPTO_ALLOW PMS_LP_MM_HP_PSRAM_ALLOW PMS_LP_MM_HP_FLASH_ALLOW PMS_LP_MM_HP_AXI_PDMA_ALLOW PMS_LP_MM_HP_KEY_MANAGER_ALLOW PMS_LP_MM_HP_DMA2D_ALLOW PMS_LP_MM_HP_PPA_ALLOW PMS_LP_MM_HP_JPEG_ALLOW (reserved) PMS_LP_MM_HP_AHB_PDMA_ALLOW PMS_LP_MM_HP_SDMMC_ALLOW PMS_LP_MM_HP_GDMA_ALLOW PMS_LP_MM_HP_USBOTG1_WRAP_ALLOW PMS_LP_MM_HP_USBOTG1_ALLOW PMS_LP_MM_HP_USBOTG_ALLOW																															
31	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

PMS_LP_MM_HP_USBOTG_ALLOW Configures whether the LP CPU in machine mode has permission to access HP high-speed USB 2.0 OTG.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_USBOTG1_ALLOW Configures whether the LP CPU in machine mode has permission to access HP full-speed USB 2.0 OTG.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_USBOTG1_WRAP_ALLOW Configures whether the LP CPU in machine mode has permission to access HP full-speed USB 2.0 OTG's wrap.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_GDMA_ALLOW Configures whether the LP CPU in machine mode has permission to access HP VDMA.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_SDMMC_ALLOW Configures whether the LP CPU in machine mode has permission to access HP SDMMC.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_AHB_PDMA_ALLOW Configures whether the LP CPU in machine mode has permission to access GDMA-AHB.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.72. PMS_LP_MM_PMS_REG1_REG (0x0030)

Continued from the previous page...

PMS_LP_MM_HP_JPEG_ALLOW Configures whether the LP CPU in machine mode has permission to access HP JPEG.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_PPA_ALLOW Configures whether the LP CPU in machine mode has permission to access HP PPA.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_DMA2D_ALLOW Configures whether the LP CPU in machine mode has permission to access HP 2D-DMA.

0: Not allowed 1: Allowed

(R/W)

PMS_LP_MM_HP_KEY_MANAGER_ALLOW Configures whether the LP CPU in machine mode has permission to access HP Key Manager.

0: Not allowed 1: Allowed

(R/W)

PMS_LP_MM_HP_AXI_PDMA_ALLOW Configures whether the LP CPU in machine mode has permission to access HP GDMA-AXI.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_FLASH_ALLOW Configures whether the LP CPU in machine mode has permission to access HP flash MSPI controller.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_PSRAM_ALLOW Configures whether the LP CPU in machine mode has permission to access HP PSRAM MSPI controller.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_CRYPTO_ALLOW Configures whether the LP CPU in machine mode has permission to access HP CRYPTO.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.72. PMS_LP_MM_PMS_REG1_REG (0x0030)

Continued from the previous page...

PMS_LP_MM_HP_GMAC_ALLOW Configures whether the LP CPU in machine mode has permission to access HP EMAC.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_USB_PHY_ALLOW Configures whether the LP CPU in machine mode has permission to access HP high-speed USB 2.0 OTG PHY.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_CSI_HOST_ALLOW Configures whether the LP CPU in machine mode has permission to access HP MIPI CSI host.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_DSI_HOST_ALLOW Configures whether the LP CPU in machine mode has permission to access HP MIPI DSI host.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_ISP_ALLOW Configures whether the LP CPU in machine mode has permission to access HP ISP.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_H264_CORE_ALLOW Configures whether the LP CPU in machine mode has permission to access HP H264 Encoder.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_RMT_ALLOW Configures whether the LP CPU in machine mode has permission to access HP RMT.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.72. PMS_LP_MM_PMS_REG1_REG (0x0030)

Continued from the previous page...

PMS_LP_MM_HP_BITSCRAMBLER_ALLOW Configures whether the LP CPU in machine mode has permission to access HP bit scrambler.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_AXI_ICM_ALLOW Configures whether the LP CPU in machine mode has permission to access HP AXI ICM.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_PERI_PMS_ALLOW Configures whether the LP CPU in machine mode has permission to access HP_PERI_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP2HP_PERI_PMS_ALLOW Configures whether the LP CPU in machine mode has permission to access LP2HP_PERI_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_DMA_PMS_ALLOW Configures whether the LP CPU in machine mode has permission to access HP_DMA_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_H264_DMA2D_ALLOW Configures whether the LP CPU in machine mode has permission to access 2D-DMA.

0: Not allowed

1: Allowed

(R/W)

Register 19.73. PMS_LP_MM_PMS_REG2_REG (0x00A4)

[illegible]

PMS_LP_MM_HP_MCPWMO_ALLOW Configures whether the LP CPU in machine mode has permission to access HP MCPWMO.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_MCPWM1_ALLOW Configures whether the LP CPU in machine mode has permission to access HP MCPWM1.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_TIMER_GROUP0_ALLOW	Configures whether the LP CPU in machine mode has permission to access HP timer group0.
--	---

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_TIMER_GROUP1_ALLOW	Configures whether the LP CPU in machine mode has permission to access HP timer group1.
--	---

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_I2C0_ALLOW Configures whether the LP CPU in machine mode has permission to access HP I2C0.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_I2C1_ALLOW Configures whether the LP CPU in machine mode has permission to access HP I2C1.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.73. PMS_LP_MM_PMS_REG2_REG (0x00A4)

Continued from the previous page...

PMS_LP_MM_HP_I2S0_ALLOW Configures whether the LP CPU in machine mode has permission to access HP I2S0.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_I2S1_ALLOW Configures whether the LP CPU in machine mode has permission to access HP I2S1.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_I2S2_ALLOW Configures whether the LP CPU in machine mode has permission to access HP I2S2.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_PCNT_ALLOW Configures whether the LP CPU in machine mode has permission to access HP PCNT.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_UART0_ALLOW Configures whether the LP CPU in machine mode has permission to access HP UART0.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_UART1_ALLOW Configures whether the LP CPU in machine mode has permission to access HP UART1.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_UART2_ALLOW Configures whether the LP CPU in machine mode has permission to access HP UART2.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.73. PMS_LP_MM_PMS_REG2_REG (0x00A4)

Continued from the previous page...

PMS_LP_MM_HP_UART3_ALLOW Configures whether the LP CPU in machine mode has permission to access HP UART3.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_UART4_ALLOW Configures whether the LP CPU in machine mode has permission to access HP UART4.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_PARLIO_ALLOW Configures whether the LP CPU in machine mode has permission to access HP PARLIO.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_GPSPI2_ALLOW Configures whether the LP CPU in machine mode has permission to access HP GP-SPI2.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_GPSPI3_ALLOW Configures whether the LP CPU in machine mode has permission to access HP GP-SPI3.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_USBDEVICE_ALLOW Configures whether the LP CPU in machine mode has permission to access HP USB/Serial JTAG Controller.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_LEDC_ALLOW Configures whether the LP CPU in machine mode has permission to access HP LEDC.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.73. PMS_LP_MM_PMS_REG2_REG (0x00A4)

Continued from the previous page...

PMS_LP_MM_HP_ETM_ALLOW Configures whether the LP CPU in machine mode has permission to access HP ETM.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_INTRMTX_ALLOW Configures whether the LP CPU in machine mode has permission to access HP interrupt matrix.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_TWAIO_ALLOW Configures whether the LP CPU in machine mode has permission to access HP TWAIO.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_TWA11_ALLOW Configures whether the LP CPU in machine mode has permission to access HP TWA11.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_TWA12_ALLOW Configures whether the LP CPU in machine mode has permission to access HP TWA12.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_I3C_MST_ALLOW Configures whether the LP CPU in machine mode has permission to access HP I3C master controller.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_I3C_SLV_ALLOW Configures whether the LP CPU in machine mode has permission to access HP I3C slave controller.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.73. PMS_LP_MM_PMS_REG2_REG (0x00A4)

Continued from the previous page...

PMS_LP_MM_HP_LCDCAM_ALLOW Configures whether the LP CPU in machine mode has permission to access HP LCD_CAM.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_ADC_ALLOW Configures whether the LP CPU in machine mode has permission to access HP ADC.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP_UHCI_ALLOW Configures whether the LP CPU in machine mode has permission to access HP UHCI.

0: Not allowed

1: Allowed

(R/W)

Register 19.74. PMS_LP_MM_PMS_REG3_REG (0x011C)

(reserved)																										PMS_LP_MM_HP_CLKRST_ALLOW						
																										PMS_LP_MM_HP_SYS_REG_ALLOW						
																										PMS_LP_MM_HP_SYSTIMER_ALLOW						
																										PMS_LP_MM_HP_IOMUX_ALLOW						
																										PMS_LP_MM_HP_GPIO_ALLOW						
31																					5	4	3	2	1	0						
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	Reset				

PMS_LP_MM_HP_GPIO_ALLOW Configures whether the LP CPU in machine mode has permission to access HP GPIO Matrix.
 0: Not allowed
 1: Allowed
 (R/W)

PMS_LP_MM_HP_IOMUX_ALLOW Configures whether the LP CPU in machine mode has permission to access HP IO MUX.
 0: Not allowed
 1: Allowed
 (R/W)

PMS_LP_MM_HP_SYSTIMER_ALLOW Configures whether the LP CPU in machine mode has permission to access HP system timer.
 0: Not allowed
 1: Allowed
 (R/W)

PMS_LP_MM_HP_SYS_REG_ALLOW Configures whether the LP CPU in machine mode has permission to access HP system register.
 0: Not allowed
 1: Allowed
 (R/W)

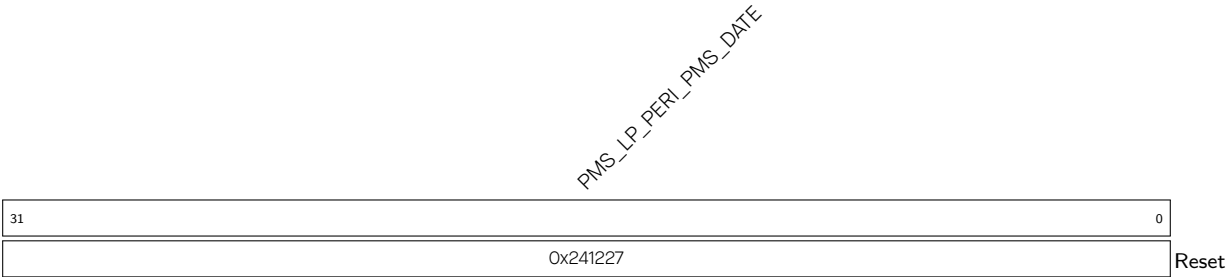
PMS_LP_MM_HP_CLKRST_ALLOW Configures whether the LP CPU in machine mode has permission to access HP_SYS_CLKRST.
 0: Not allowed
 1: Allowed
 (R/W)

19.7.4 LP_PERI_PMS_REG

The addresses in this section are relative to the LP_PERI_PMS base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

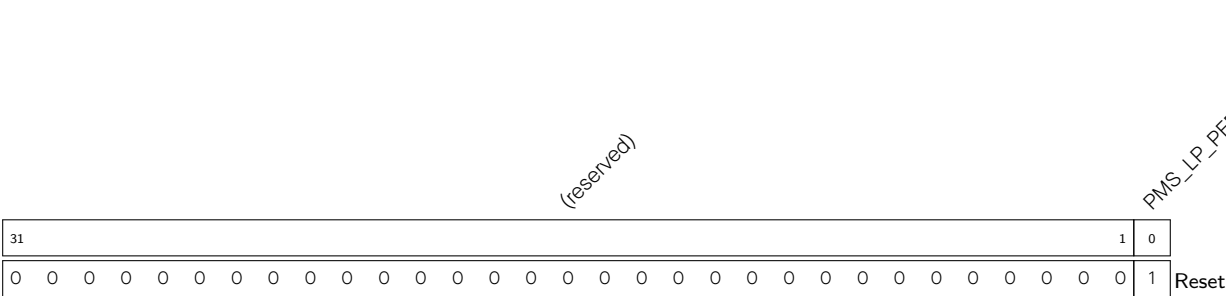
For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 19.75. PMS_LP_PERI_PMS_DATE_REG (0x0000)



PMS_LP_PERI_PMS_DATE Version control register (R/W)

Register 19.76. PMS_LP_PERI_PMS_CLK_EN_REG (0x0004)



PMS_LP_PERI_PMS_CLK_EN Configures whether to keep the clock always on.

- 0: Enable automatic clock gating
- 1: Keep the clock always on

(R/W)

Register 19.77. PMS_LP_MM_LP_PERI_PMS_REGO_REG (0x0008)

(reserved)							PMS_LP_MM_LP_TRNG_ALLOW		(reserved)		PMS_LP_MM_LP_HUK_ALLOW		PMS_LP_MM_LP_TSNS_ALLOW		PMS_LP_MM_LP_HP2LP_PMS_ALLOW		PMS_LP_MM_LP_EFUSE_ALLOW		PMS_LP_MM_LP_INTR_ALLOW		PMS_LP_MM_LP_IOMUX_ALLOW		PMS_LP_MM_LP_TOUCH_ALLOW		PMS_LP_MM_LP_ADC_ALLOW		PMS_LP_MM_LP_I2S_ALLOW		PMS_LP_MM_LP_I2CMST_ALLOW		(reserved)		PMS_LP_MM_LP_UART_ALLOW		PMS_LP_MM_LP_PERICLKST_ALLOW		PMS_LP_MM_LP_MAILBOX_ALLOW		PMS_LP_MM_LP_WDT_ALLOW		PMS_LP_MM_LP_PMU_ALLOW		PMS_LP_MM_LP_ANAPERI_ALLOW		PMS_LP_MM_LP_AONCLKST_ALLOW		PMS_LP_MM_LP_SYSREG_ALLOW	
31	25						24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Reset																
0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

PMS_LP_MM_LP_SYSREG_ALLOW Configures whether LP CPU in machine mode has permission to access LP system registers.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_AONCLKRST_ALLOW Configures whether LP CPU in machine mode has permission to access LP_AONCLKRST (LP always-on clock and reset).

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_TIMER_ALLOW Configures whether LP CPU in machine mode has permission to access LP timer.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_ANAPERI_ALLOW Configures whether LP CPU in machine mode has permission to access LP ANAPERI (analog peripherals).

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_PMU_ALLOW Configures whether LP CPU in machine mode has permission to access LP PMU (Power Management Unit).

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_WDT_ALLOW Configures whether LP CPU in machine mode has permission to access LP WDT (watchdog timer).

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.77. PMS_LP_MM_LP_PERI_PMS_REGO_REG (0x0008)

Continued from the previous page...

PMS_LP_MM_LP_MAILBOX_ALLOW Configures whether LP CPU in machine mode has permission to access LP Mailbox Controller.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_PERICLK_RST_ALLOW Configures whether LP CPU in machine mode has permission to access LP PERICLK_RST (peripheral clock and reset).

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_UART_ALLOW Configures whether LP CPU in machine mode has permission to access LP UART.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_I2C_ALLOW Configures whether LP CPU in machine mode has permission to access LP I2S.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_SPI_ALLOW Configures whether LP CPU in machine mode has permission to access LP SPI.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_I2CMST_ALLOW Configures whether LP CPU in machine mode has permission to access LP I2C master.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_I2S_ALLOW Configures whether LP CPU in machine mode has permission to access LP I2S.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.77. PMS_LP_MM_LP_PERI_PMS_REGO_REG (0x0008)

Continued from the previous page...

PMS_LP_MM_LP_ADC_ALLOW Configures whether LP CPU in machine mode has permission to access LP ADC.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_TOUCH_ALLOW Configures whether LP CPU in machine mode has permission to access LP touch sensor.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_IOMUX_ALLOW Configures whether LP CPU in machine mode has permission to access LP IO MUX.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_INTR_ALLOW Configures whether LP CPU in machine mode has permission to access LP INTR (interrupt).

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_EFUSE_ALLOW Configures whether LP CPU in machine mode has permission to access LP eFuse.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_LP_PMS_ALLOW Configures whether LP CPU in machine mode has permission to access LP_PERI_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

PMS_LP_MM_HP2LP_PMS_ALLOW Configures whether LP CPU in machine mode has permission to access HP2LP_PERI_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.77. PMS_LP_MM_LP_PERI_PMS_REGO_REG (0x0008)

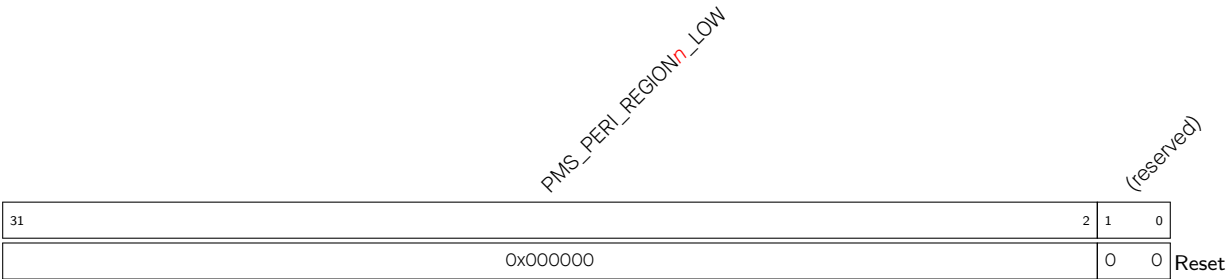
Continued from the previous page...

PMS_LP_MM_LP_TSENS_ALLOW Configures whether LP CPU in machine mode has permission to access LP temperature sensor.
0: Not allowed
1: Allowed
(R/W)

PMS_LP_MM_LP_HUK_ALLOW Configures whether LP CPU in machine mode has permission to access LP HUK (Hardware Unique Key).
0: Not allowed
1: Allowed
(R/W)

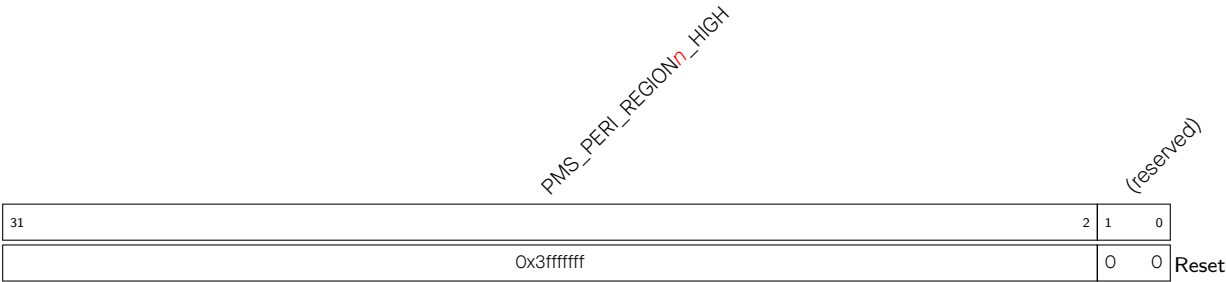
PMS_LP_MM_LP_TRNG_ALLOW Configures whether LP CPU in machine mode has permission to access LP TRNG.
0: Not allowed
1: Allowed
(R/W)

Register 19.78. PMS_PERI_REGION n _LOW_REG (n : 0-1) (0x000C+0x8* n)



PMS_PERI_REGION n _LOW Configures the high 30 bits of the start address of peripheral register's region n . (R/W)

Register 19.79. PMS_PERI_REGION n _HIGH_REG (n : 0-1) (0x0010+0x8* n)



PMS_PERI_REGION n _HIGH Configures the high 30 bits of the end address of peripheral register’s region n . (R/W)

Register 19.80. PMS_PERI_REGION_PMS_REG (0x001C)

(reserved)																						PMS_HP_CORE1_MM_REGION_PMS					PMS_HP_CORE1_UM_REGION_PMS					PMS_HP_CORE0_MM_REGION_PMS					PMS_HP_CORE0_UM_REGION_PMS					PMS_LP_CORE_REGION_PMS				
31										10										9	8	7	6	5	4	3	2	1	0	Reset																
0										0										3		3		3		3		3																		

PMS_LP_CORE_REGION_PMS Configures whether LP core in machine mode has permission to access address region0 and address region1. Bit0 corresponds to region0 and bit1 corresponds to region1.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE0_UM_REGION_PMS Configures whether HP CPU0 in user mode has permission to access address region0 and address region1. Bit2 corresponds to region0 and bit3 corresponds to region1.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE0_MM_REGION_PMS Configures whether HP CPU0 in machine mode has permission to access address region0 and address region1. Bit4 corresponds to region0 and bit5 corresponds to region1.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE1_UM_REGION_PMS Configures whether HP CPU1 in user mode has permission to access address region0 and address region1. Bit6 corresponds to region0 and bit7 corresponds to region1.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE1_MM_REGION_PMS Configures whether HP CPU1 in machine mode has permission to access address region0 and address region1. Bit8 corresponds to region0 and bit9 corresponds to region1.

0: Not allowed

1: Allowed

(R/W)

Register 19.81. PMS_PERI_REGION_2_TO_7_PMS_REG (0x0020)

(reserved)		PMS_HP_CORE1_MM_REGION_2_TO_7_PMS				PMS_HP_CORE1_UM_REGION_2_TO_7_PMS				PMS_HP_CORE0_MM_REGION_2_TO_7_PMS				PMS_HP_CORE0_UM_REGION_2_TO_7_PMS				PMS_LP_CORE_REGION_2_TO_7_PMS			
31	30	29	24	23	18	17	12	11	6	5	0										
0	0	0x3F		0x3F		0x3F		0x3F		0x3F		Reset									

PMS_LP_CORE_REGION_2_TO_7_PMS Configures whether LP CPU in machine mode has permission to access address region2 to address region7. Bit0 corresponds to region2 and bit5 corresponds to region7.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE0_UM_REGION_2_TO_7_PMS Configures whether HP CPU0 in user mode has permission to access address region2 to address region7. Bit0 corresponds to region2 and bit5 corresponds to region7.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE0_MM_REGION_2_TO_7_PMS Configures whether HP CPU0 in machine mode has permission to access address region2 to address region7. Bit0 corresponds to region2 and bit5 corresponds to region7.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE1_UM_REGION_2_TO_7_PMS Configures whether HP CPU1 in user mode has permission to access address region2 to address region7. Bit0 corresponds to region2 and bit5 corresponds to region7.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE1_MM_REGION_2_TO_7_PMS Configures whether HP CPU0 in machine mode has permission to access address region2 to address region7. Bit0 corresponds to region2 and bit5 corresponds to region7.

0: Not allowed

1: Allowed

(R/W)

Register 19.82. PMS_PERI_REGION m _LOW_REG (m : 2-7) (0x0024+0x8*($m-2$))

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PMS_PERI_REGION m _LOW Configures the high 30 bits of the start address of peripheral register's region m . (R/W)

Register 19.83. PMS_PERI_REGION m _HIGH_REG (m : 2-7) (0x0028+0x8*($m-2$))

			PMS_PERI_REGION_HIGH		(reserved)	
31	2	1	0			
0x3ffffff			0	0	Reset	

PMS_PERI_REGION m _HIGH Configures the high 30 bits of the end address of peripheral register's region m . (R/W)

19.7.5 HP2LP_PERI_PMS_REG

The addresses in this section are relative to the H2LP_PERI_PMS base address provided in Table 7.3-2 in Chapter 7 [System and Memory](#).

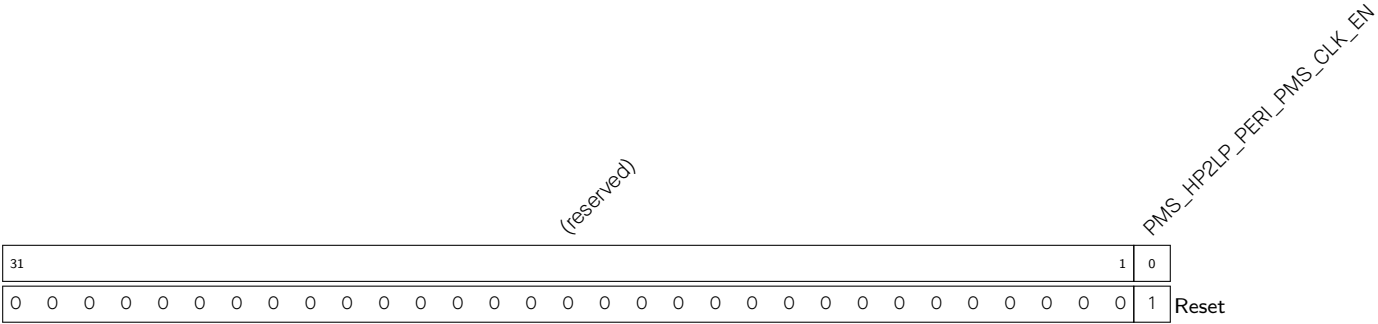
For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 19.84. PMS_HP2LP_PERI_PMS_DATE_REG (0x0000)

31		0	
0x230406		Reset	

PMS_HP2LP_PERI_PMS_DATE Version control register (R/W)

Register 19.85. PMS_HP2LP_PERI_PMS_CLK_EN_REG (0x0004)



PMS_HP2LP_PERI_PMS_CLK_EN Configures whether to keep the clock always on.

- 0: Enable automatic clock gating
- 1: Keep the clock always on

(R/W)

Register 19.86. PMS_HP_CORE n _MM_PMS_REGO_REG (n : 0-1) (0x0008+0x8* n)

(reserved)								PMS_HP_COREn_MM_LP_TRNG_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_SRAM_ALLOW																							
(reserved)								PMS_HP_COREn_UM_LP_HUK_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_TSNS_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_PMS_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_EFUSE_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_INTR_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_IOMUX_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_TOUCH_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_ADC_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_I2S_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_I2CMST_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_SPI_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_I2C_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_UART_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_PERICKRST_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_MAILBOX_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_WDT_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_PMU_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_ANAPER_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_TIMER_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_AONCLKRST_ALLOW																							
(reserved)								PMS_HP_COREn_MM_LP_SYSREG_ALLOW																							

31								25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	Reset	

PMS_HP_CORE_n_MM_LP_SYSREG_ALLOW Configures whether HP CPU_n in machine mode has permission to access LP System Registers.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE_n_MM_LP_AONCLKRST_ALLOW Configures whether HP CPU_n in machine mode has permission to access LP_AONCLKRST.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE_n_MM_LP_TIMER_ALLOW Configures whether HP CPU_n in machine mode has permission to access LP timer.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE_n_MM_LP_ANAPERI_ALLOW Configures whether HP CPU_n in machine mode has permission to access LP ANAPERI.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_PMU_ALLOW Configures whether HP CPU n in machine mode has permission to access LP PMU.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.86. PMS_HP_CORE n _MM_PMS_REGO_REG (n : 0-1) (0x0008+0x8* n)

Continued from the previous page...

PMS_HP_CORE n _MM_LP_WDT_ALLOW Configures whether HP CPU n in machine mode has permission to access LP WDT.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_MAILBOX_ALLOW Configures whether HP CPU n in machine mode has permission to access LP Mailbox Controller.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_PERICKRST_ALLOW Configures whether HP CPU n in machine mode has permission to access LP PERICKRST.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_UART_ALLOW Configures whether HP CPU n in machine mode has permission to access LP UART.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_I2C_ALLOW Configures whether HP CPU n in machine mode has permission to access LP I2C.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_SPI_ALLOW Configures whether HP CPU n in machine mode has permission to access LP SPI.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_I2CMST_ALLOW Configures whether HP CPU n in machine mode has permission to access LP I2C master.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.86. PMS_HP_CORE n _MM_PMS_REGO_REG (n : 0-1) (0x0008+0x8* n)

Continued from the previous page...

PMS_HP_CORE n _MM_LP_I2S_ALLOW Configures whether HP CPU n in machine mode has permission to access LP I2S.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_ADC_ALLOW Configures whether HP CPU n in machine mode has permission to access LP ADC.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_TOUCH_ALLOW Configures whether HP CPU n in machine mode has permission to access LP touch sensor.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_IOMUX_ALLOW Configures whether HP CPU n in machine mode has permission to access LP IO MUX.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_INTR_ALLOW Configures whether HP CPU n in machine mode has permission to access LP INTR.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_EFUSE_ALLOW Configures whether HP CPU n in machine mode has permission to access LP eFuse.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_PMS_ALLOW Configures whether HP CPU n in machine mode has permission to access LP_PERI_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.86. PMS_HP_CORE n _MM_PMS_REGO_REG (n : 0-1) (0x0008+0x8* n)

Continued from the previous page...

PMS_HP_CORE n _MM_HP2LP_PMS_ALLOW Configures whether HP CPU n in machine mode has permission to access HP2LP_PERI_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_TSENS_ALLOW Configures whether HP CPU n in machine mode has permission to access LP temperature sensor.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_SRAM_ALLOW Configures whether HP CPU n in machine mode has permission to access LP SRAM.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _MM_LP_TRNG_ALLOW Configures whether HP CPU n in machine mode has permission to access LP TRNG.

0: Not allowed

1: Allowed

(R/W)

Register 19.87. PMS_HP_CORE n _UM_PMS_REGO_REG (n : 0-1) (0x000C+0x8* n)

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PMS_HP_CORE_n_UM_LP_SYSREG_ALLOW Configures whether HP CPU_n in user mode has permission to access LP System Registers.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE_n_UM_LP_AONCLKRST_ALLOW Configures whether HP CPU_n in user mode has permission to access LP_AONCLKRST.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_TIMER_ALLOW Configures whether HP CPU n in user mode has permission to access LP timer.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE_n_UM_LP_ANAPERI_ALLOW Configures whether HP CPU_n in user mode has permission to access LP ANAPERI.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_PMU_ALLOW Configures whether HP CPU n in user mode has permission to access LP PMU.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.87. PMS_HP_CORE n _UM_PMS_REGO_REG (n : 0-1) (0x000C+0x8* n)

Continued from the previous page...

PMS_HP_CORE n _UM_LP_WDT_ALLOW Configures whether HP CPU n in user mode has permission to access LP WDT.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_MAILBOX_ALLOW Configures whether HP CPU n in user mode has permission to access LP Mailbox Controller.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_PERICKRST_ALLOW Configures whether HP CPU n in user mode has permission to access LP PERICKRST.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_UART_ALLOW Configures whether HP CPU n in user mode has permission to access LP UART.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_I2C_ALLOW Configures whether HP CPU n in user mode has permission to access LP I2C.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_SPI_ALLOW Configures whether HP CPU n in user mode has permission to access LP SPI.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_I2CMST_ALLOW Configures whether HP CPU n in user mode has permission to access LP I2C master.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.87. PMS_HP_CORE n _UM_PMS_REGO_REG (n : 0-1) (0x000C+0x8* n)

Continued from the previous page...

PMS_HP_CORE n _UM_LP_I2S_ALLOW Configures whether HP CPU n in user mode has permission to access LP I2S.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_ADC_ALLOW Configures whether HP CPU n in user mode has permission to access LP ADC.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_TOUCH_ALLOW Configures whether HP CPU n in user mode has permission to access LP touch sensor.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_IOMUX_ALLOW Configures whether HP CPU n in user mode has permission to access LP IO MUX.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_INTR_ALLOW Configures whether HP CPU n in user mode has permission to access LP INTR.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_EFUSE_ALLOW Configures whether HP CPU n in user mode has permission to access LP eFuse.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_PMS_ALLOW Configures whether HP CPU n in user mode has permission to access LP_PERI_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

Continued on the next page...

Register 19.87. PMS_HP_CORE n _UM_PMS_REGO_REG (n : 0-1) (0x000C+0x8* n)

Continued from the previous page...

PMS_HP_CORE n _UM_HP2LP_PMS_ALLOW Configures whether HP CPU n in user mode has permission to access HP2LP_PERI_PMS_REG.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_TSENS_ALLOW Configures whether HP CPU n in user mode has permission to access LP temperature sensor.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_HUK_ALLOW Configures whether HP CPU n in user mode has permission to access LP HUK.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_SRAM_ALLOW Configures whether HP CPU n in user mode has permission to access LP SRAM.

0: Not allowed

1: Allowed

(R/W)

PMS_HP_CORE n _UM_LP_TRNG_ALLOW Configures whether HP CPU n in user mode has permission to access LP TRNG.

0: Not allowed

1: Allowed

(R/W)